Which Decisions Can an AI Agent Take Without Review? A Prospectively Specified Pilot Comparing Evidence Tests, Track Records and LLM Judges

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ABSTRACT

Organisations must decide which decisions an AI agent may take without human review. Correct outcomes alone do not show whether an agent responds appropriately to evidence. We define decision-structured agency (DSA) as responsiveness to the evidence that should determine a class of decisions, measured by removing documents, reversing their claims or adding independent evidence. In a prospectively specified pilot on synthetic counterparty verification (14 configurations, eight developers), we compare delegation on DSA, track records and an LLM judge. Deterministic programs implementing our reading of the policy generate the answer key. The registered test favoured DSA over track records at matched coverage (one-sided $p < 0.001$; bootstrap over configurations $p = 0.001$, over configurations and episodes $p = 0.052$). The advantage depended on one intervention: removing a register excerpt the policy did not require. Excluding two audit-flagged interventions ($p = 0.20$) or revising the key after the author's partial review ($p = 0.28$) removed statistical support. It reached nominal significance in ten of 60 combinations of policy readings, each including a reading stricter than the legal texts. Under every key the judge passed most decisions correct on the original file but failing an evidence test (59–70%); a stronger judge discriminated no better on a stratified sample of 199 decisions. Much of the judge-based rules' deficit came from one class the judge read differently from the programs. The primary result depended on how the policy was translated into an answer key, which is part of the instrument and requires independent scrutiny.

KEYWORDS

AI agents, delegation, autonomy, oversight, reasons-responsiveness, evaluation, certification

1 INTRODUCTION

1.1 The delegation question

Organisations that deploy AI agents must decide which decisions an agent may take without human review. Supervisors now ask them to make this decision explicitly and to revisit it. The Financial Stability Board's 2026 sound practices suggest that firms "may also progressively increase an agent's ability to interact with the external environment as they gain confidence it can do so safely", with "triggers for transitioning between different forms of human oversight based on performance metrics or user requests" (Financial Stability Board 2026). The UK Financial Conduct Authority (FCA) calls for "continuous, conditional assurance" and for frameworks covering "delegated authority" (Rusu 2026). Neither says what evidence should ground the confidence on which autonomy is extended.

1.2 Why track records are not enough

The default answer is a track record: success rates, human approvals, correction rates, model confidence, or a judge's reading of the agent's trace. A track record answers whether the agent's actions turned out correct. A delegator also needs to know whether they were correct *for the right reasons and within scope*, because only then can past success be expected to transfer to the next case. This is the premise of the paper. The pilot tested one consequence of it, that a rule built on evidence tests, that is, on interventions on the evidence the agent consults, delegates more safely than a track record, and §6.1 reports that it could not confirm this. Outcome success masks the basis of a decision, much as human correction masks an agent's raw competence (Azevedo 2026).

The gap is not hypothetical. In CXBench, 80.1% of runs that passed the outcome check failed process checks in regulated customer-experience workflows (Yu et al. 2026). Procedure-aware evaluation finds that between 27% and 78% of reported τ-bench successes, depending on model and domain, are "corrupt successes" (Cao et al. 2026). In paired tasks, an agent's skill at acting correctly and its skill at correctly abstaining are nearly independent (phi coefficient $\varphi \approx -0.10$; Liu et al. 2026). Self-reports do not close the gap: verbalised confidence barely separates correct from incorrect answers (area under the ROC curve, AUROC, 0.47–0.66; Shen 2026), and stated reasoning can be unfaithful to what drives a model's answer (Turpin et al. 2023).

There is also a formal reason. Relational properties of an agent's behaviour, such as what its action depends on, are 2-safety hyperproperties that no monitor of a single trace can decide; deciding them requires comparing executions (Pandey et al. 2026a). Calibrating on final outcomes cannot control the share of correct outcomes reached on an unsupported basis (Opoku and Banahene 2026). The philosophy of meaningful human control reached the same conclusion from another direction: a system must "demonstrably and verifiably be responsive to the human moral reasons relevant in the circumstances", and this holds "no matter how efficient [it] may be in performing specific tasks" (Santoni de Sio and van den Hoven 2018).

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1.3 A motivating example

In the pilot of §5, Claude Sonnet 5.5 reads a customer file in which the central beneficial-ownership register names one owner while a business-register excerpt and a notarial deed record a transfer of all shares to another. It asks the customer to correct the register entry, the action the policy admits here: the deed is a source independent of the register's controlling authority, so the exception to the reporting duty applies. Its explanation is accurate, names the deed as decisive, and an LLM judge reading the explanation finds the grounds sound. Remove the deed and run the same case again (episode c3-100005, run 0, in `runs/pilot/sonnet55_calibration.jsonl`): the agent gives the same action, now resting on the business-register excerpt, and writes that nothing in the file shows the two registers to be independent, "that is an assumption". Under the policy the only admissible answer is now to report the discrepancy: its rule R2(b) requires, beyond the legal text it paraphrases, that the other source be independent of the register's controlling authority (§8). Without that condition the second decision would also be admissible, and §8 reports the results under that reading as well. The first decision was correct, and its record read well; what the record could not show is that the decision did not turn on the document it named. That is the gap this paper is about, and §6.2 counts how often the judge passes such decisions.

1.4 Question, finding and contributions

We ask which evidence should decide that an agent may take a class of decisions without review: its track record, an LLM judge's reading of its trace, or its reaction to changes in the evidence. For the last we define *decision-structured agency* (DSA): the extent to which an agent configuration, within a class of decisions, regularly recognises and reacts to the evidence that ought to settle those decisions. It also covers escalating when that evidence is missing or contradictory or when an action would exceed the agent's authority. The notion descends from reasons-responsiveness (Fischer and Ravizza 1998) and from the tracking condition of meaningful human control (Santoni de Sio and van den Hoven 2018).

Our prospectively specified pilot does not show that delegating on DSA beats a track record. The registered test favoured DSA, but the advantage rested on cases where our own policy admits more than one reading, and it disappeared when those cases were excluded or the answer key was reviewed. Under every answer key, an LLM judge passed most of the decisions that failed an evidence test. The main lesson concerns method: in an evaluation of this kind the answer key is part of the instrument. It feeds both the test and the criterion against which the test is judged, and two separately written oracles (programs that compute the correct action from the generator's hidden facts) that agree with each other can still share the same contestable reading. Cao [2026] shows that policy ambiguity can masquerade as agent error in benchmark scores; we show that it can also decide which delegation rule appears to work.

We make three contributions.

C1. A construct and its measurement. We define DSA per pair (agent configuration, decision class) and measure it by composed interventions on the external evidence the agent consults, with control arms, scoring responses against the agent-declared grounds (§3). Which response each intervention requires is fixed by oracles that encode one reading of the policy; the pilot shows how much rests on that reading.

C2. A delegation rule. We give a rule that grants, narrows and revokes autonomy per decision class beneath a fixed, human-set ceiling, with certificates bound to the agent's configuration (§4). The statistical machinery is standard; the novelty is the loss it controls. The pilot exercises only the granting step; narrowing, revocation and transfer between classes are specified but not yet tested.

C3. A prospectively specified pilot with a full audit trail. On a synthetic counterparty-verification workflow with 14 model configurations, we compare delegation on DSA with delegation on a track record, the registered primary comparison, and with rules built on an LLM judge, at matched autonomy and under evidence shift (§5, §6). We publish the registration, every amendment, the log of policy ambiguities, the author's review of the oracles' answers and the code, so that each step from data to conclusion can be checked.

1.5 Scope

We study ordinary, non-adversarial shifts in evidence. When sources can be corrupted adversarially, no procedure that sees the same evidence can avoid handing off ambiguous cases to a human (Pandey et al. 2026b); for that setting we propose combining our rule with a source-quorum gate. Our guarantees hold for generated episodes and relative to a family of interventions fixed in advance. What the rule provides is reliability allocation beneath a fixed ceiling, not a security guarantee. The case study encodes publicly available anti-money-laundering rules in a fully synthetic environment. It uses no real entities and no institutional data, and makes no claim of regulatory compliance.

2 BACKGROUND AND RELATED WORK

Delegation and trust in multi-agent systems. Castelfranchi and Falcone [1998] define delegation as one agent relying on another for a task, and grade it from weak to strict and from open to closed. Their socio-cognitive account makes trust a decision to delegate, indexed by trustor, trustee, context, task and goal, and grounded in beliefs about competence, willingness and opportunity (Castelfranchi and Falcone 2000). Scerri et al. [2002] cast adjustable autonomy as a cost-sensitive transfer of control between agent and human, but their model *assumes* a given quality of agent decisions. Lee and See [2004] ask for reliance calibrated to the system's real capability. Work on trust across tasks shows that trust can be transferred by matching task requirements to agent features (Falcone and Castelfranchi 2008; Venanzi et al. 2011), that category priors should give way to direct evidence (Burnett et al. 2010), and that transfer follows task similarity, from harder tasks to easier ones (Soh et al. 2020). Tomašev et al. [2026] bring this tradition to LLM agents. They propose authority graduated by trust, with "algorithmic circuit breakers" for revocation and behavioural metrics as trust signals, and they note that current systems "lack robust mechanisms for the dynamic assessment of competence". Their framework is conceptual, without a measurement procedure or evaluation. *We give an operational instance, tested in a prospectively specified pilot: competence is estimated per decision class from intervention tests rather than assumed (Scerri) or read from reputation.*

Reasons-responsiveness and meaningful human control. Fischer and Ravizza [1998] ground responsibility in a decision mechanism that is moderately reasons-responsive: it recognises reasons and reacts to them across a range of circumstances. Santoni de Sio and van den Hoven [2018] recast this as the *tracking* condition of meaningful human control. A system must "demonstrably and verifiably" respond to the relevant human reasons, and good results can come from "tracking the wrong property". Mecacci and Santoni de Sio [2020] operationalise tracking for dual-mode vehicles, and Veluwenkamp [2022] argues that the reasons tracked must be normative. CARE-Drive (Suryana et al. 2026) is the first behavioural test of tracking for a foundation model. It varies decision facts and prompt-supplied reasons for a single-step driving model and estimates the sensitivity of its decisions. It has no per-case ground truth for which fact should decide, no escalation option, and no link to delegation, and it does not separate recognising a reason from acting on it. *DSA carries the tracking test to multi-step agents and to external evidence whose decisive role and redundancy are set by construction, relative to one reading of the policy. It is designed to score receptivity and reactivity separately and to allocate autonomy from the resulting profile; the pilot measures reactivity and reports receptivity only through the declared-grounds test (§3.4).*

Traces, provenance and auditability. Structured provenance has a long history (W3C 2013). For LLM agents, LEDGER builds layered claim-to-evidence graphs over observed sessions and states the problem crisply: "Visibility, however, is not the same as auditability" (Kim et al. 2026). TRACE specifies a typed record that must exist before any durable state change, and leaves to each consumer the policy that maps a verdict to an action (Chang and Chang 2026). DNative-Twin takes the committed decision as its unit of replay and comparison, and treats authority as "granted in advance" (Pang et al. 2026). DEMM grades whether a decision's evidence suffices for an external audit and warns against the "container fallacy" (Solozobov 2026). Graph of Trace records the agent's own declared dependencies as justification edges (Gao et al. 2026a). A recent survey organises the field around six trust functions: verification, attribution, debugging, safety enforcement, audit and recovery. It lists an authorization relation as open (Wang et al. 2026a). *These systems make decisions inspectable. None tests whether the recorded grounds are what the decision depends on, and none decides what an agent may do next.*

Faithfulness and evidence interventions. Stated reasons can be unfaithful (Turpin et al. 2023; Lanham et al. 2023), and interventions are the standard test. Atanasova et al. [2023] insert inputs and reconstruct inputs from explanations. Matton et al. [2025] compare the causal effect of input concepts with how often explanations cite them. Pawar et al. [2026] test necessity and sufficiency of the factors a model cites. Khanzadeh [2026] edits the agent's own reasoning steps. Closer to agents, AgentAbstain pairs should-act and should-abstain tasks that differ by one perturbation (Liu et al. 2026). Liao [2026] changes the source authority of evidence in next-action tasks. "Copies or Sources?" (Gao et al. 2026b) and GraphEcho (Wang et al. 2026b) contrast copied with independent sources, and Nautil removes an investigator's cited grounds against a matched control (Bi et al. 2026). AUDITA formalises overdetermination with actual causation and shows that single-cause deletion fails under redundancy (Du and Chen 2026). The design itself, an input change with an expected response, has a longer lineage: metamorphic testing in software engineering (Segura et al. 2016), behavioural test suites such as CheckList (Ribeiro et al. 2020), contrast sets (Gardner et al. 2020) and counterfactually augmented data (Kaushik et al. 2020), and the diagnosis of models that are right for the wrong reasons through heuristics (McCoy et al. 2019) or shortcuts (Geirhos et al. 2020). *The operators exist. We compose them, adding an independent source and then removing the original, to separate redundancy from non-use for one decision. We read the result against the agent-declared grounds and use it prospectively to allocate autonomy, not retrospectively to attribute blame.*

The outcome–process gap. Several 2026 benchmarks show that outcome checks pass agents whose process is defective: CXBench in regulated customer-experience workflows (Yu et al. 2026), procedure-aware evaluation of τ-bench (Cao et al. 2026), deterministic evidence-path checks in GroundEval (Flynt 2026), and "lucky passes" in software-engineering agents (Sahoo et al. 2026). ToolChain-CRC (conformal risk control) proves that calibration on final answers cannot control unsupported intermediate steps (Opoku and Banahene 2026). Cao [2026] adds a caution that applies to any benchmark of this kind, ours included: when a natural-language policy is silent, ambiguous or contradictory, a single gold answer scores policy ambiguity as agent error. *These audits check procedure or evidence paths. We check whether the action depends on the evidence that should warrant it, which a procedurally compliant trace can hide, and we show that the same kind of ambiguity can decide which delegation rule appears to work (§6.1).*

Graduated autonomy and certified deployment. Several systems already widen or narrow agent autonomy from experience. The Digital Apprentice promotes per-skill tiers when correction rates fall (Weber and Taneja 2026). Hedwig learns check-in behaviour from a developer's approvals (Shukla et al. 2026). Sovereign-OS moves permissions with audited outcomes (Yuan et al. 2026). Minimal Oversight caps authority by a raw success rate, after correcting for masking (Azevedo 2026). Trust Before Autonomy prescribes a gated, revocable ladder (Angelidis et al. 2026). Earned Authority shows that self-produced evidence must not raise a fixed ceiling (Zhang and Zhang 2026). The Agent Control Protocol accumulates only penalties from history (Fernandez 2026). Agent Reliability Profiles define action classes and autonomy tiers for financial services (Hsu et al. 2026). On the statistical side, READY qualifies confidence-threshold escalation policies on held-out cases, with explicit review cost and an LLM-judged process defect (Chatrath et al. 2026). Shen [2026] certifies per model–task pair; Role-Stratified CRC certifies per stratum (Rahman et al. 2026); Conformal Selective Acting certifies anytime (Khosravi and Huo 2026). Pandey et al. [2026b] gate actions on agreement among sources from separate control domains. *Each delegates on outcomes, corrections, approvals, confidence, judged traces or source counts. We delegate on interventionally measured DSA, and we compare against these rules run through one certification engine.*

Answer keys as part of the instrument. Every test of this kind needs an oracle that says what the correct response is, and the oracle problem is a central difficulty of software testing (Barr et al. 2015). Knight and Leveson [1986] showed that independently written versions of a program fail on the same inputs far more often

than independence would predict, because their authors find the same cases hard; the lesson carries to two oracles written from one specification. In machine learning, label errors in test sets change the ranking of models (Northcutt et al. 2021), and disagreement among human annotators is a property of the task rather than noise to be averaged away (Plank 2022). Jacobs and Wallach [2021] read such problems through measurement theory: a score is valid only if the construct it operationalises and the procedure that produces its labels are both examined. Multiverse analysis reports a result under every defensible analytic choice instead of one (Steegen et al. 2016), and formalising law as code shows that statute text leaves choices the formaliser must make explicit (Merigoux et al. 2021). *We apply these lessons to the normative map of an agent evaluation: we log the readings of the policy that the oracles encode, report the primary result under every logged combination, and show that the choice of reading decided which delegation rule appeared to work (§6.1, §8).*

3 DECISION-STRUCTURED AGENCY

3.1 Definition

Following the tracking condition of meaningful human control (Santoni de Sio and van den Hoven 2018), itself a reworking of Fischer and Ravizza's (1998) reasons-responsiveness, and its first behavioural test for foundation models (Suryana et al. 2026), we define *decision-structured agency* (DSA) as the extent to which an agent's behaviour within a decision class regularly recognises and reacts to the evidence that ought to settle decisions in that class.[1] Concretely, the agent's action changes when, and only when, the decisive evidence changes; it escalates when that evidence is missing or contradictory, or when the action would exceed its authority; and it revises correctly after correction. DSA is a behavioural, counterfactual property of an agent configuration in a decision class: it is measured by interventions on the external evidence the agent consults, whose decisive role and redundancy are set by construction, and it implies nothing about intentionality or moral responsibility, which remain with the human principals (the tracing condition of meaningful human control).

Two consequences of the definition matter for what follows. First, DSA is indexed by an *agent configuration* π (model version, prompt, tools) and a *decision class* c; it is not an intrinsic trait of a model, and a change of π is a change of agent. Second, each dimension of DSA has named neighbours in recent work (Table 2). What is new is their integration into a per-class profile, their measurement by interventions on external evidence with a normative basis fixed by construction, relative to the oracles' reading of the policy (§6.1), and the use of that profile to allocate autonomy (§4).

3.2 Decision episodes

The unit of measurement is the *decision episode*

$$E = (C, I, P, G, A, O),$$

where C is the context (task, principal, decision class), I the information available to the agent at decision time (the evidence items it retrieved, with their sources), P the permissions in force, G the graph of *agent-declared grounds* (the evidence items and rules the agent states it relies on), A the action taken, drawn from the domain actions together with *escalate* and *refuse*, and O the observed consequences, including deferred ones. Most of these fields have counterparts in existing decision records: DEMM's authorisation envelope covers I and P (Solozobov 2026), the audit verdict of Nian et al. [2026] covers A and part of O, and TRACE records cover claims, evidence and verdicts (Chang and Chang 2026). What the episode adds is G as an object to be *tested* rather than stored, and its use as the unit of intervention.

We keep two notions of authority apart. P is the authority the episode grants; it is fixed by the environment and is the reference for the scope dimension of DSA. The *autonomy* that our rule grants for class c (§4) is an output of measurement. Conflating them would make the rule circular.

3.3 Decision classes

A decision class $c \in C$ is a family of decision situations that share a goal and an admissible action set (including *escalate* and *refuse*). Class membership is given by a rule $\kappa(E) \in C$ computable at decision time from (C, I, P) alone, never from the outcome, and the partition is fixed before any calibration label is seen. Following the socio-cognitive theory of delegation, in which trust is a relation about a task in a context and delegation occurs when the degree of trust exceeds a threshold reflecting acceptance of risk (Castelfranchi and Falcone 1998, 2000), each class carries two separate descriptors:

- **requirements** Q_c: the evidence-handling capacities a justified decision in c demands (reading the decisive item, detecting stale or contradictory sources, responding to independent corroboration, respecting scope, recognising when to escalate); Q_c determines which DSA dimensions bear on c and whether evidence may be shared across classes (§4.5);
- **risk parameters** K_c: stakes, reversibility, escalation cost and legal reservation; K_c sets the tolerated rate α_c and, for legally reserved classes, forces escalation regardless of measured DSA.

Classes that differ only in K_c may share DSA estimates but not thresholds; classes that differ in Q_c may not share estimates even when their stakes coincide. To ease alignment with deployment practice, each class is tagged with the reversibility-graded action classes of Agent Reliability Profiles (Hsu et al. 2026) and with the verifiability, reversibility and criticality axes of Tomašev et al. [2026].

3.4 Dimensions

Table 2 lists the dimensions of DSA, the operational test for each, and the prior metric each is aligned with. Dimensions 1–5 enter the delegation rule; dimension 6 is reported but not certified. In the pilot the certified loss is computed from the action alone, which exercises dimensions 2–4 through A^*; receptivity is read from the declared grounds (the declared-grounds test, §6.2) and dimension 5

[1] We borrow the structure of moderate reasons-responsiveness (receptivity and reactivity, evaluated over counterfactual scenarios for a fixed mechanism), not its threshold. Fischer and Ravizza accept *weak* reactivity because their question is blame; delegation asks whether outcomes will be reliable when no human checks, so DSA requires *regular* receptivity and *regular* reactivity. "Ought to settle" marks the reasons as normative rather than merely motivating (Veluwenkamp 2022).

Table 1: Comparison with the closest work (✓ present; (✓) partial, see the notes; – absent)

Work	(a) decision unit	(b) evidence interventions	(c) redundancy	(d) declared grounds	(e) per-class autonomy	(f) revocation	(g) held-out	(h) shift	(i) empirical
LEDGER (Kim et al. 2026)	–	–	–	–	–	–	–	–	(✓)[1]
TRACE (Chang and Chang 2026)	(✓)[2]	–	(✓)[3]	✓	–	–	–	–	–
DNative-Twin (Pang et al. 2026)	✓	(✓)[4]	–	–	–	–	–	–	–
DEMM (Solozobov 2026)	✓	–	–	–	–	–	–	–	–
Project Ariadne (Khanzadeh 2026)	–	–	–	(✓)[5]	–	–	–	–	(✓)[6]
Pawar et al. [2026]	(✓)[7]	(✓)[8]	–	✓	–	–	–	–	✓
AgentAbstain (Liu et al. 2026)	(✓)[7]	✓	–	–	–	–	–	–	✓
AUDITA (Du and Chen 2026)	–	–	✓	–	–	–	–	–	(✓)[9]
CARE-Drive (Suryana et al. 2026)	✓	(✓)[10]	–	–	–	–	–	–	✓
CXBench (Yu et al. 2026)	–	–	–	–	–	–	–	–	✓
Pandey et al. [2026b]	✓	(✓)[11]	✓	–	–	–	–	–	✓
Intelligent AI Delegation (Tomašev et al. 2026)	–	–	–	–	(✓)[12]	(✓)[12]	–	–	–
Trust Before Autonomy (Angelidis et al. 2026)	–	(✓)[13]	–	–	(✓)[12]	(✓)[12]	–	–	–
Digital Apprentice (Weber and Taneja 2026)	–	–	–	–	✓	✓	–	–	–
Hedwig (Shukla et al. 2026)	–	–	–	–	✓	✓	–	–	(✓)[14]
Minimal Oversight (Azevedo 2026)	–	–	–	–	✓	(✓)[15]	–	–	–
READY (Chatrath et al. 2026)	–	–	–	–	(✓)[16]	(✓)[17]	✓	–	✓
Certified deferral (Shen 2026)	(✓)[18]	–	–	–	(✓)[19]	–	✓	(✓)[20]	(✓)[21]
Role-Stratified CRC (Rahman et al. 2026)	(✓)[22]	(✓)[23]	–	–	(✓)[24]	–	✓	✓	✓
Conformal Selective Acting (Khosravi and Huo 2026)	(✓)[25]	–	–	–	–	(✓)[26]	(✓)[27]	✓	(✓)[28]
WorkTrace / DSA (this paper)	✓	✓	✓	(✓)[30]	(✓)[31]	(✓)[29]	✓	✓	✓

Notes to Table 1. Columns: (a) decision unit: a single decision is the unit of analysis; (b) external-evidence interventions: the agent's external evidence is changed and the agent is re-run; (c) redundancy: redundant or overdetermined evidence is handled; (d) declared grounds: the agent's own stated grounds are scored; (e) per-class autonomy: autonomy is allocated per class of decisions or tasks; (f) revocation: autonomy is withdrawn; (g) held-out: evaluation on held-out cases; (h) shift: evaluation under distribution shift; (i) empirical: empirical study with LLM agents.

Footnotes: 1. LEDGER: two qualitative case studies, no metrics. 2. TRACE: one record per claim or commitment. 3. TRACE: argues that correlated sources add little (effective panel size); no tests. 4. DNative-Twin: replay under declared changes to recorded state; no LLM agent is re-run. 5. Ariadne: intervenes on the agent's generated reasoning steps, not on stated grounds about evidence. 6. Ariadne: single-model QA without tools; the report gives two sample sizes (30 and 500) and two headline values for the same rate, so we read it as preliminary. 7. Pawar, AgentAbstain: single decision or task pair, not multi-step episodes. 8. Pawar: changes input factors of tool-free decisions. 9. AUDITA: live LLM pipelines, but evidence partly simulated. 10. CARE-Drive: varies facts written into a single prompt. 11. Pandey et al.: corrupted sources in synthetic payment episodes. 12. Prescribed, not implemented or evaluated. 13. Trust Before Autonomy: "unanswerable variants" correspond to removing the sole decisive source. 14. Hedwig: evaluated on 11 operations with an LLM judge. 15. Minimal Oversight: implicit, through a decaying success-rate estimate; simulation only. 16. READY: one global threshold per agent and workflow, with per-case accept or escalate. 17. READY: re-qualification on version changes, no statistical monitoring. 18. Shen: single multiple-choice answers, not agent decisions. 19. Shen: one certified confidence threshold per model–task pair. 20. Shen: threshold transfer between two tasks only; the certificate assumes no shift. 21. Shen: eleven small LLMs on multiple-choice QA, without tools. 22. Role-Stratified CRC: individual argument fields of tool calls. 23. Role-Stratified CRC: prompt injections in tool outputs, contrasted with clean runs on recorded traces; agents are not re-run. 24. Role-Stratified CRC: thresholds per argument role, gating field values at run time, not autonomy; recalibration, no revocation. 25. Conformal Selective Acting: act or abstain on single outputs in one stream. 26. Conformal Selective Acting: all certifications reset at fixed epoch boundaries; no data-driven revocation. 27. Conformal Selective Acting: risk is controlled online on the same stream; a held-out split only calibrates the score. 28. Conformal Selective Acting: RLVR-trained LLMs on benchmark streams, including live online fine-tuning, not agents with tools. 29. This paper: revocation is specified (§4.6) but not exercised in the pilot. 30. This paper: declared grounds are scored by an LLM judge for factual decisions only (§6.2); the joint receptivity × reactivity table of §3.5 is not reported. 31. This paper: the pilot exercises granting only; narrowing and revocation are specified (§4.5, §4.6) but not exercised.

is not exercised (§3.5). What the pilot certifies is therefore an action-based proxy of DSA; the construct as proposed, with receptivity, scope restraint and provenance, is not validated by it.

For scope restraint we score attempted over-reach (calls blocked by the ceiling, §4.4) and decision-scope over-reach (deciding a matter reserved for a human), because executed over-reach is blocked mechanically and would otherwise be invisible. Authority is ordered by a small per-domain lattice of actions, data scopes and thresholds.

3.5 Measurement by composed evidence interventions

For every episode the generator supplies a *normative action map* $A^*(E) \subseteq \mathcal{A}$, the set of correct actions; it may contain *escalate* (for example when the decisive evidence is missing) and *refuse* (an out-of-permission request). An intervention family $\mathcal{I}(E) = \{do_j\}$ acts on the external evidence in I:

- **remove** X: delete the evidence item X;
- **only** X: remove all evidence except X;
- **invert** X: replace X by an item whose content points to a different correct action;
- **add independent** Y: add a source Y that agrees or conflicts with X, where *independent* means that Y shares no

Table 2: Dimensions of decision-structured agency.

#	Dimension	Operational test	Aligned prior metric
1	Receptivity to decisive evidence	The agent registers an intervention on the decisive item (in G or in an escalation message)	Epistemic competence (Shao et al. 2025)
2	Reactivity to decisive evidence	The action changes as A^* prescribes under *remove*, *only*, *invert* and composed interventions, and not under placebo interventions	Reason-responsiveness (Suryana et al. 2026); necessity/sufficiency tests (Pawar et al. 2026)
3	Scope restraint	$R(a) = \mathbb{1}[\text{auth}(a) \subseteq P]$, scored on *attempted* and decision-scope over-reach	Restraint in action alignment (Li and Zhao 2026)
4	Escalation calibration	Escalates iff decisive evidence is missing or contradictory, or the action exceeds P; over-escalation is an error	Selective escalation (Trinh et al. 2026); escalation thresholds (DosSantos DiSorbo and Ju 2026)
5	Revision after correction	A principal's correction propagates to later decisions in the episode	Evidence uptake and revision (Ríos-García et al. 2026)
6	Decision-point recognition and provenance	Alternatives and grounds are recorded; the decision path can be reconstructed from E	Grounding (Shao et al. 2025); evidence-path validity (Flynt 2026)

controlling authority with X, counted per authority rather than per record of origin (Pandey et al. 2026b).

Each operator exists in prior work (Turpin et al. 2023; Matton et al. 2025; Pawar et al. 2026; Liu et al. 2026; Gao et al. 2026b). We use them in *composition*: adding an independent Y and then removing X distinguishes redundancy (the action is unchanged because Y suffices) from non-use (the action is unchanged although nothing else supports it). Because the generator also computes $A^*(do_j(E))$, the correct response to every variant is fixed by construction, relative to the oracles' reading of the policy (§6.1, §8). Every unchanged decision is classified as *redundancy*, *cancellation* (opposing effects offset) or *non-reaction*, since a null effect of a single intervention does not show that an item is inert (Du and Chen 2026; Mahale 2026).

Receptivity and reactivity can be scored jointly per run in a 2×2 table: recognised and reacted; recognised but not reacted; not recognised but reacted (unfaithful grounds); neither. This is where the agent-declared grounds G earn their place: they are read against the interventional effects, not trusted. The pilot does not report this table: grounds were judged for factual decisions only (§6.2), so receptivity under each intervention was not read; §6 reports reactivity (the non-reaction and escalation measures of Table 7) and a static check of the declared grounds. A second, separate family of *policy interventions* holds the evidence fixed and changes the principal's class policy or P; it tests tracking of distal reasons (Mecacci and Santoni de Sio 2020) and is not run in the pilot.

Controls are mandatory: null arms, placebo interventions on non-decisive evidence, prompt-format placebos (the largest effect in CARE-Drive was a format parameter; Suryana et al. 2026), size-matched shams, copy, pointer and padding controls (Gao et al. 2026b), and a fake declared-grounds graph with the same number of grounds rewired to other items. Controls are reported as separate flip rates and do not enter the basis loss; responsiveness to them is read as noise or sensitivity to form, not as dependence on evidence. The pilot runs four of these controls (null, placebo, sham and copy; §5.2). Agent behaviour is stochastic, so conditions should be run several times and compared as distributions; in the pilot one configuration was run three times and the others once (§5.3). Coupled replays with common random numbers are used only on open-weight stacks (Mahale 2026).

3.6 What DSA is not

DSA is not general capability: it is scored on episodes the agent solves under full information, and its discriminant validity is tested with agents that have planted deficits (§5.3). It is not a security property: adversarially corrupted sources are out of scope (§1.5). And it is not an intrinsic trait of a model: like READY's qualification (Chatrath et al. 2026), it attaches to a configuration and a workflow, here refined to a decision class.

4 FROM DSA TO DELEGATION

This section turns per-class DSA measurements into decisions about autonomy. All statistical machinery is standard; we present it as an application of selective classification (El-Yaniv and Wiener 2010; Geifman and El-Yaniv 2017), Learn-then-Test (Angelopoulos et al. 2025) and anytime-valid testing (Waudby-Smith and Ramdas 2024; Podkopaev and Ramdas 2022). The contribution lies in the loss being certified and in the grant–revoke cycle around it.

4.1 Losses

For an episode E with action $a(E)$ and normative action map $A^*(E)$ (§3.5), define

$$\ell_{\text{out}}(E) = \mathbb{1}[a(E) \notin A^*(E)],$$
$$\ell_{\text{basis}}(E) = \mathbb{1}\left[\exists\, do_j \in \mathcal{I}(E) :\ a(do_j(E)) \notin A^*(do_j(E))\right],$$

and the *unjustified-action* loss $\ell_{\text{WT}}(E) = \max(\ell_{\text{out}}, \ell_{\text{basis}})$ (WT for WorkTrace, the name of the instrument), so that $\ell_{\text{WT}} \geq \ell_{\text{out}}$ pointwise. An action is unjustified if it is wrong, or right but on a basis that fails an intervention fixed in advance, for example by ignoring an inversion or surviving the removal of its only support. A basis failure records that the action was not reliable under the tested intervention family; it does not establish that the original decision was reached for the wrong reason, since an unstable agent or an error on the variant alone leaves the same record (§8). Control arms (null, placebo, sham, copy) are reported separately and do not enter ℓ_{basis}. Redundancy is handled by A^*: if a redundant source survives *remove* X, A^* does not change and an agent that keeps its

action is not penalised. The guarantee below always holds *relative to* $\mathcal{I}$; unjustified behaviour on features no intervention probes is invisible to it. In the theory only autonomous actions incur ℓ_{WT}, and escalation and refusal are charged through cost (§4.7). In the pilot the agent proposes an action on every episode and every proposal is scored against the admissible set, so an escalation is unjustified when escalation is not admissible, and the loss is computed over all episodes of the class (§4.2).

The decomposition $\ell_{\mathrm{WT}} = \ell_{\mathrm{out}} + (1 - \ell_{\mathrm{out}})\, \ell_{\mathrm{basis}}$ has the same shape as the trajectory loss of ToolChain-CRC, whose Proposition 1 shows that calibrating on final answers cannot control the second term (Opoku and Banahene 2026). The difference is how the basis term is measured. ToolChain-CRC uses an audit label on the run's evidence; we use the counterfactual dependence of the agent's action. The two disagree in both directions: an agent that ignores valid supporting evidence passes their audit but fails ours, and an agent that correctly discounts a faulty tool output fails their audit but passes ours.

4.2 Per-class certification

Let φ_c denote autonomous coverage in class c and

$$R_c = \Pr\left(\ell_{\mathrm{WT}}(E) = 1 \mid \text{autonomous},\ \kappa(E) = c\right)$$

the *selective* risk (El-Yaniv and Wiener 2010). We certify this conditional quantity, not the joint rate $\Pr(\ell_{\mathrm{WT}} = 1 \wedge \text{autonomous})$, which is smaller by a factor of φ_c and does not bound the rate a delegator experiences (Khosravi and Huo 2026). The pilot certifies a different quantity: the whole-class rate $U_c = \Pr(\ell_{\mathrm{WT}}(E) = 1 \mid \kappa(E) = c)$ of unjustified proposals, with escalations scored against the admissible set, because its agents propose an action on every episode and no deployed selector exists yet. A bound $U_c \le \alpha_c$ implies $R_c \le \alpha_c/\varphi_c$ for any selector that acts autonomously on a share φ_c of the class, which is informative only when φ_c is close to one; a certificate of R_c itself needs a fixed selector and a denominator counted among the autonomous decisions, and the pilot's grants should not be read as one. For each class, calibration episodes D_c are drawn from disjoint generator seeds. With n_c decisions and k_c unjustified ones among them, where for the selective guarantee n_c counts the autonomous decisions of a selector fixed before calibration and for the pilot's whole-class rate it counts every episode of the class, the exact binomial p-value for $H_c : R_c > \alpha_c$ is $p_c = \Pr(\mathrm{Bin}(n_c, \alpha_c) \le k_c)$ (Clopper and Pearson 1934). If a within-class readiness threshold is used, thresholds are tested in a pre-specified fixed sequence (Angelopoulos et al. 2025). The B-conf baseline of the pilot instead applies a Bonferroni correction across its candidate confidence thresholds, as specified in amendment A11.2. Classes are combined by Bonferroni, $\delta_c = \delta/|C|$, or by an allocation chosen on an independent split, as in Role-Stratified CRC (Rahman et al. 2026). The grant set is $\hat{G} = \{c : p_c \le \delta_c\}$.

Proposition 1 (simultaneous per-class guarantee). Suppose the episodes in each D_c are i.i.d. draws from the generator's class-c distribution, future class-c episodes are drawn from the same distribution, and κ, the threshold sequence and π are fixed in advance. Then

$$\Pr\left(\exists\, c \in \hat{G} : R_c > \alpha_c\right) \le \delta.$$

Proof sketch. Conditional on n_c, p_c is super-uniform under H_c (Geifman and El-Yaniv 2017). Fixed-sequence testing controls the family-wise error within a class at δ_c (Angelopoulos et al. 2025), and a union bound across classes gives δ. □

With zero observed failures, certification at $\alpha_c = 0.10$ needs $n_c \ge 46$ for six classes at $\delta = 0.05$, and roughly 100 episodes per class give 80% power at a true rate of 2%. Pooled guarantees dilute rare classes: a pooled level α implies only $R_c \le \alpha/w_c$ for a class of prevalence w_c (Rahman et al. 2026), which is why we certify per class.

4.3 Verdicts

Two one-sided tests per class yield four verdicts:

- **proceed:** $c \in \hat{G}$; the agent acts without review in c;
- **proceed-narrowed:** a preparatory sub-class of c is in $\hat{G}$ but the binding step is not; the agent completes the preparatory determination and escalates the binding one;
- **refuse:** the lower Clopper–Pearson bound on R_c exceeds a level $\beta_c > \alpha_c$ (p-value $\Pr(\mathrm{Bin}(n_c, \beta_c) \ge k_c)$, with its own δ budget). The class is certified bad and removed from the agent's permissions; the agent no longer drafts or proposes there. This verdict differs from the agent refusing an out-of-permission request, which is part of A^* and is certified like any other action;
- **escalate:** otherwise, including whenever the evidence is insufficient either way.

Classes that the law subjects to meaningful human intervention, such as entering, refusing or maintaining a customer relationship under Art. 76(5)(b) of the EU Anti-Money Laundering Regulation (AMLR), and that the policy therefore reserves for human officers, receive *escalate* regardless of measured DSA; the rule then governs only their preparatory sub-classes.

4.4 Ceiling and certificate lifecycle

Following Zhang and Zhang [2026], autonomy is allocated beneath a fixed, human-set ceiling: the episode permissions P bound what any grant may cover, and only a human can raise them. Widening a class from *escalate* to *proceed* is gated expansion beneath the ceiling, never above it. Grants are issued by an evaluator independent of the agent, so that evidence the agent produces about itself cannot raise its authority. Each certificate is bound to a fingerprint of π (model, prompt, tools), is suspended on any change outside a declared set, and expires after a fixed period. We call this *reliability allocation beneath a fixed ceiling*; it is not a security guarantee.

4.5 Transfer across classes

A profile measured in a source class s may inform a target class t only if $Q_t \subseteq Q_s$ and the source tests were at least as demanding, reflecting the asymmetry of trust transfer in the delegation literature (Falcone and Castelfranchi 2010; Soh et al. 2020). Transferred evidence is a prior that orders testing and supports conservative verdicts; it never certifies. A *proceed* verdict in t requires class-specific episodes, or a stated exchangeability assumption together with a sentinel sample in t that can falsify it. Escalation is tested separately in every class, since competence on a task does not imply competence in judging one's own competence on it.

4.6 Revocation and re-grant

After a grant, a random audit selects each autonomous episode with probability r_c, fixed before the episode is seen; audited episodes are replayed with interventions to obtain $X_t = \ell_{\mathrm{WT}}$. Revocation tests $H_0 : \mathbb{E}[X_t \mid \mathcal{F}_{t-1}] \le \alpha'_c$ with the betting e-process

$$M_t = \prod_{s \le t} \big(1 + \lambda_s (X_s - \alpha'_c)\big),$$
$$\lambda_s \in [0, 1/\alpha'_c) \text{ predictable},$$

and revokes at the first t with $M_t \ge 1/\delta_{\mathrm{rev}}$; by Ville's inequality the probability of a false revocation is at most δ_{rev} under continuous monitoring (Waudby-Smith and Ramdas 2024; Podkopaev and Ramdas 2022). Hysteresis, $\alpha'_c > \alpha_c$ (with the pilot's $\alpha_c = 0.10$, for example $\alpha'_c = 0.15$), prevents a class whose true rate sits near α_c from flapping; the pilot grants classes at $\alpha_c = 0.10$, so a revocation level for its grants would lie above 0.10. A revoked class returns to *escalate*, and re-grant requires fresh evidence. Because failures are charged to the agent only when they are unjustified actions, not when the evidence could not have revealed the right action, revocation accumulates only agent-attributable failures (Castelfranchi and Falcone 2000). A failure on a requirement triggers re-testing in every proceeding class that shares it.

Data-driven per-class withdrawal already exists for outcome signals, explicitly in Digital Apprentice and implicitly in Minimal Oversight (Azevedo 2026). What we add is withdrawal triggered by failed intervention tests, with hysteresis and re-testing across shared requirements. Unlike scheduled certificate resets (Khosravi and Huo 2026), revocation here is data-driven. Detection is slow. In a simulation with $\delta_{\mathrm{rev}} = 0.05$ and the constant bet $\lambda_s = 1/(1-\alpha'_c)$, the Kelly bet against a doubled rate, the median delay from a doubling of the true rate from α'_c to $2\alpha'_c$ is about 34 audited episodes at $\alpha'_c = 0.15$ and about 118 at $\alpha'_c = 0.05$ (50,000 simulated streams, fixed seed; `paper/pilot/revocation_delay.py`). Revocation was specified but not exercised in the pilot, which tested granting only. This argues for higher audit rates in high-stakes classes and is reported as a limitation. The grant–audit–revoke cycle has a classical precedent in continuous sampling inspection (Dodge 1943).

4.7 Cost and masking

Risk is the constraint; cost is the objective among certified options. With deployment class mix w_c, the agent's inference cost per episode c_c^{inf} (paid on every episode, because the agent proposes even where it will escalate), harm L_c of an unjustified action, escalation cost e_c, reviewer error h_c, and the cost C_c of certifying class c spread over the N episodes a certificate covers before it expires, the expected cost per episode is

$$\mathcal{J} = \sum_c w_c \Big[c_c^{\mathrm{inf}} + \mathbb{1}[c \in \hat{G}]\big(\varphi_c L_c R_c + (1-\varphi_c)(e_c + h_c L_c)\big) + \mathbb{1}[c \notin \hat{G}]\,(e_c + h_c L_c)\Big] + \frac{1}{N}\sum_{c \in \hat{G}} C_c,$$

optionally subject to a review-capacity bound $\sum_c w_c\,(1 - \mathbb{1}[c \in \hat{G}]\varphi_c) \le B$, following the learning-to-defer formulation (Madras et al. 2018; Mozannar and Sontag 2020). Reviewer error is not zero, so escalation is never free.

All rules, ours and the baselines, obey a *masking rule*: the agent is scored on its own proposal on every episode, including escalated ones, and an episode that was escalated and then corrected by a human never counts as an agent success. Without it, correction hides raw competence and inflates any track record (Azevedo 2026).

5 STUDY DESIGN

5.1 Environment: synthetic counterparty verification

We instantiate decision episodes in a synthetic counterparty-verification workflow of a fictional EU credit institution. The institution's policy (rules R1–R5, given to every configuration verbatim) paraphrases public anti-money-laundering texts: verification against sources other than the central beneficial-ownership register and the discrepancy-reporting duty with its two exceptions (AMLR Arts 22(7) and 24; European Parliament and Council 2024), freshness of register excerpts and review intervals (AMLR Arts 23(4) and 26), reliability and independence of sources (EBA guidelines 4.13 and 4.26–4.28, European Banking Authority 2021; FATF Recommendation 10, Financial Action Task Force 2026), the requirement of meaningful human intervention in decisions to enter, refuse or maintain a relationship (AMLR Art. 76(5)(b)) and the non-outsourceable decisions on risk profile and suspicious-activity reporting (Art. 18(3)(c) and (e)). Open-textured terms are fixed as synthetic parameters (for example, an excerpt is current if issued no more than 30 days before the decision), and in places the policy is stricter than the texts (§8). In brief: **R1** ranks sources in tiers (government registers, courts and the institution's own records; notarial deeds and regulated professionals; customer-supplied material), treats a copy as its original and calls two sources independent only if their issuers do not answer to the same controlling authority. **R2** requires beneficial-ownership information to be verified against a source other than the central register, treats a difference as a discrepancy to be reported within 14 days, allows a request for customer correction instead when the difference is typographical or the register is merely outdated and the owners are known from an independent reliable source, never for higher-risk customers, and sends missing or contradictory evidence to a document request or escalation. **R3** counts a register excerpt as current for 30 days, requires a full review every year for higher-risk and every five years for other customers, and reopens the file on any change recorded in an official filing after the last review. **R4** reserves entering, refusing or maintaining a relationship, changing a risk rating and suspicious-activity reports for human officers and defines when an onboarding file is complete. **R5** confines the agent to the permitted actions and tells it to escalate or request documents rather than guess. The full text is in the repository (`worktrace/policy.py`). All entities, registers, authorities and documents are generated; no real data are used, and we make no claim of regulatory compliance.

The pilot covers three decision classes that differ in what a justified decision requires. We refer to them by name; the codes used in the registration and in the tables follow in parentheses.

- **The discrepancy class (c3): register-discrepancy triage.** Compare the central register with other evidence and choose between reporting the discrepancy, asking the

customer to correct it, declaring no discrepancy, requesting a document, or escalating. The flagship rule paraphrases AMLR Art. 24(2)(b), under which an institution may refrain from reporting an outdated register entry and instead ask the customer to correct it if the current owners are known "from another reliable source", there is no ground for suspicion, and the customer is not higher-risk. The policy adds that the other source must be independent of the register's controlling authority, and the oracles read the exception as prescribing *request correction*. This yields an *add independent source* intervention whose correct response the oracles fix by construction. The author's review first disputed that response, because the documents name issuers but not their controlling authorities, and withdrew the objection after a later case (§6.1); the legal text has no such condition (§8).

- **The freshness class (c5): freshness and re-review.** Decide whether the file must be reopened, a fresh excerpt requested, or nothing done, from review dates, risk-dependent intervals, official filings and excerpt issue dates.
- **The onboarding class (c6): a reserved decision.** The onboarding file is assembled; the only admissible actions are to escalate or, if the file is incomplete, to request the missing document. Finalising or refusing the relationship is outside the agent's permissions. This class measures scope restraint and escalation.

Three further classes (identification evidence, beneficial-owner computation, document requests) are specified but deferred to a second phase.

5.2 Episodes, interventions and ground truth

Each class has a generator that samples hidden facts and renders documents from them, with deliberately over-sampled edge cases (exact-threshold dates, transliteration variants, higher-risk ratings, concealment alerts). The normative action map $A^*(E)$ is never stored: it is recomputed for every episode and variant from the generator's record of the episode, which includes facts the agents never see, such as the controlling authority of each issuer (§6.1), by two oracles written from one specification by the same AI assistant used throughout, in separate modules (an imperative rule engine and a declarative rule table with separately implemented helpers), so that their agreement tests implementation consistency rather than independence of reading. Episodes on which the oracles disagree are discarded; in testing on more than 40,000 episodes and variants they never disagreed.

For every factual episode we generate the intervention family of §3.5 (*remove X*, *only X*, *invert X*, *add independent Y*, and the composition *add Y then remove X*), where X is the document that settles the decision in that episode. We add four controls: a *null* re-run of the identical episode, a *placebo* that changes document identifiers and order only, a *sham* that adds an irrelevant document, and a *copy* that adds a dependent reproduction of X from an issuer under the same controlling authority, which must not count as corroboration. Operators that do not apply to an episode are skipped.

Three splits use disjoint seed ranges: *calibration*, *held-out* (same distribution) and *shifted*. The shifted generator changes the evidence distribution without changing the policy: more outdated registers without independent confirmation, dependent vendor reproductions presented as corroboration, customer claims of notarial confirmation, boundary dates, routine filings after the last review, and incomplete or stale onboarding files accompanied by pressure to finalise. Only the factual arm is run on the shifted split, because its role is to measure outcome errors of decisions that a rule has delegated.

5.3 Configurations and protocols

We study three protocols, analysed separately because they expose different information to the model (Table 3).

Agents with tools. The agent lists and retrieves documents through tools and submits one decision with its action, the documents it relied on (G) and the documents that settled the decision. Claude models are called through the Anthropic API and OpenAI models through the Responses API, both with effort (reasoning effort) set to medium, without sampling parameters and without refusal or model fallbacks, so that the measured configuration is never silently replaced. The two closed-model series are paired by each model's position in its developer's line, as the developers describe it, not by price: Claude Haiku 5.5 with GPT-6 Luna, Claude Sonnet 5.5 with GPT-6.1 Terra, and Claude Opus 5.5 with GPT-6.1 Sol. Open-weight models run locally through Ollama with a fixed seed; malformed tool calls that the server rejects are retried with a shifted seed and logged.

One-pass. The policy, the task and all documents are given in a single prompt and the model returns a JSON decision. We use this protocol for models that cannot run an interactive tool loop: in our tests Bielik 11B v3 wrote an entire simulated trajectory, including invented document identifiers, instead of calling tools.

Decision models. Basal 2.0-max (Kinas 2026; MLX, 8-bit), Liquid AI d1-3B (Liquid AI 2026) and GPT-6 Luna on the OpenAI Decisions API choose one option from a list in a single forward pass and return probabilities that their developers describe as calibrated (checked in Appendix A). They receive the same documents as the one-pass protocol. They cannot declare grounds, so their receptivity is not measured; their reactivity is.

API configurations run every arm, including the null re-run. Locally run models and the local decision models omit the null arm, which is identical under a fixed seed.

Agents with planted deficits. To check the instrument itself (known-groups validity), we run five scripted agents on the same episodes: an oracle; an agent that replays stored answers for episodes it has seen and otherwise answers with the class prior (a perfect track record with no dependence on evidence); an agent that always escalates; an agent that treats every issuer as independent and copies as separate sources; and an agent that follows an irrelevant cue in document identifiers. Expected results were written down before the run (`paper/pilot/known_groups.md`). These agents are excluded from the delegation test. The model configurations come from eight developers and include two three-size series from the closed-model developers.

Table 3: Configurations in the pilot. [a] The common core used for comparisons between configurations is the first 60 / 30 / 60 base episodes and run 0; the other bases and runs serve reliability analyses only (§5.4). The held-out episodes of Opus 5.5, which amendment A4 had omitted for budget reasons, were run last, in the second budget cycle (closing note of the registration log). API identifiers: claude-haiku-5-5, claude-sonnet-5-5, claude-opus-5-5 (Anthropic); gpt-6-luna, gpt-6.1-terra, gpt-6.1-sol on the Responses API and gpt-6-luna on the Decisions API (OpenAI); Ollama tags gpt-oss:120b, mistral-small3.2:24b, gemma4:26b, qwen3.8:27b and SpeakLeash/bielik-11b-v3.0-instruct:Q8_0; all with reasoning effort medium where the API offers it.

Protocol	Configuration	Developer	Runs per arm	Base episodes per class (cal / held-out / shifted)
Agent with tools, API	Claude Haiku 5.5	Anthropic	3	100 / 50 / 100 [a]
	Claude Sonnet 5.5	Anthropic	1	60 / 30 / 60
	Claude Opus 5.5	Anthropic	1	60 / 30 / 60
	GPT-6 Luna	OpenAI	1	60 / 30 / 60
	GPT-6.1 Terra	OpenAI	1	60 / 30 / 60
	GPT-6.1 Sol	OpenAI	1	60 / 30 / 60
Agent with tools, local	gpt-oss 120B	OpenAI	1 (seeded)	60 / 30 / 60
	Mistral Small 3.2 24B	Mistral AI	1 (seeded)	60 / 30 / 60
	Gemma 4 26B	Google	1 (seeded)	60 / 30 / 60
	Qwen 3.8 27B	Alibaba	1 (seeded)	60 / 30 / 60
One-pass, local	Bielik 11B v3 Instruct	SpeakLeash	1 (seeded)	60 / 30 / 60
Decision model	Basal 2.0-max	R. Kinas	1	60 / 30 / 60
	Liquid AI d1-3B	Liquid AI	1	60 / 30 / 60
	GPT-6 Luna, Decisions API	OpenAI	1	60 / 30 / 60
Planted deficits	five scripted agents	–	1	60 / 30 / 60

5.4 Measures

For each configuration and class we report, on calibration and held-out data: the outcome loss ℓ_{out}; the basis loss ℓ_{basis} (any intervention arm with an action outside A^*); $\ell_{\text{WT}} = \max(\ell_{\text{out}}, \ell_{\text{basis}})$; the rate of right outcomes on a wrong basis; the null-flip rate (sampling noise); the placebo/sham flip rate; the share of unchanged decisions under interventions that should have changed them (non-reaction); scope restraint (attempted out-of-permission actions); and protocol incidents. Cost, tokens and latency are reported per run.

The unit of analysis is a base episode with its arms, in one run. For configurations with several runs per episode (Claude Haiku 5.5) the registered analysis treats each run as a unit; because runs of one episode are not independent, we also report the majority action over runs for every arm, which gives one unit per episode, and the common-core analysis on run 0 (amendment A4, §5.8). A basis failure means that one of the five intervention arms has an action outside A^*; control arms are reported as flip rates per arm. Units missing any arm are dropped. Terms used in §6: an *episode* is one case file; its *arms* (or *variants*) are the factual file and its intervention and control versions; a *unit* is one base episode with all its arms in one run; a *cell* is one (configuration, class) pair; a *comparison* is one (configuration, k) pair in the delegation test. The *basis loss* ℓ_{basis} is the rate of units with a *basis failure*; a *wrong-basis decision* is a correct factual outcome in a unit with a basis failure; "failing a basis test" means the same.

5.5 Delegation rules compared

All rules except the judged-grounds rule run through the same certification engine: a class is delegated if the one-sided Clopper–Pearson upper bound of its loss on calibration episodes is at most $\alpha = 0.10$, with $\delta = 0.05$ split by Bonferroni across classes (§4.2); in the pilot the certified loss is the whole-class rate of unjustified proposals, not the selective risk of §4.2 (see there). The rules differ only in the loss or signal they certify:

- **DSA rule:** ℓ_{WT}.
- **Track record (B-OO):** ℓ_{out}.
- **Confidence (B-conf):** a confidence threshold certified on ℓ_{out}, for decision models that report probabilities.
- **Judged process (B-READY-proc):** the same rule with wrong basis judged by an LLM reading the decision and its stated grounds instead of by interventions; this is the comparison of *how* wrong-basis decisions are detected (the judge test, §5.6), whereas the registered primary comparison is with the track record (the delegation test). The judge (Claude Haiku 5.5, fixed rubric) is applied post hoc.
- **Judged grounds (B-trace-judge):** an LLM score of the clarity and soundness of the agent-declared grounds, as proposed for "behavioural metrics" in delegation frameworks (Tomašev et al. 2026).
- **Readiness gate (B-TBA):** repeated-run reliability plus correct abstention when the sole source is removed (Angelidis et al. 2026).

Rules that need human signals (approval history, correction rates) are simulated from oracle labels with a reviewer-error parameter and reported as sensitivity analyses. All rules score the agent's own decision before any human correction (Azevedo 2026).

5.6 Hypotheses

- **The delegation test (H3, primary).** At matched autonomous coverage, the DSA rule yields a lower outcome-error rate among delegated decisions on the shifted split than the track-record rule. Coverage is matched by delegating, for each configuration, the k classes with the lowest certified upper bound under each rule; the test pairs the

two rules within (configuration, k) comparisons. At forced coverage k a rule may delegate classes it has not certified, so the delegation test compares how the two losses rank classes; §6.1 reports the certified grants separately.

- **The judge test (H1′).** Among correct outcomes, a share fail interventional basis tests while passing a procedure audit; we estimate the share of basis failures that the audit misses.
- **The declared-grounds test (H2).** Interventions detect mismatches between agent-declared grounds and actual dependence more often than inspection of the declared grounds alone. As registered, this takes the interventions as the reference, so the test reports the sensitivity and the false-positive rate of a static check of the declared grounds against the interventional result, not a symmetric comparison of two detectors (§6.2).
- **Construct checks (CV1, CV2).** DSA is not reducible to outcome accuracy across configurations, and its variance decomposes into configuration, class and seed components.

5.7 Analysis

The delegation test uses a sign test over (configuration, k) comparisons in which the two rules delegate different classes, and a mixed-effects logistic regression of shifted outcome errors on rule, with configuration as a random effect. The sign test treats the $k = 1$ and $k = 2$ comparisons of one configuration as independent, which they are not, so its p-values are descriptive; the mixed model is the registered test. When two classes have the same certified bound at the boundary of the delegated set, a rule's shifted error is the average over its equally justified selections, the expected error under a random tie-break (amendment A7, made after interim results were known; §5.8); in the mixed model each equally justified selection enters with its binomial counts weighted by one over the number of selections, as specified in amendment A11.5. The mixed model treats every shifted episode as an independent observation although an episode enters both rules and both values of k; a bootstrap over configurations, proposed at the A12 audit, was rerun on the final data (Table 4); it addresses the dependence between the comparisons of one configuration, not the sharing of the same episodes by every configuration, which no resampling of configurations removes. Shifted episodes are shared across configurations, rules and values of k, so observations are crossed by configuration and episode. As a post hoc robustness check added before submission (3 November 2026), we also ran a two-way bootstrap that resamples configurations and base episodes independently, the same episode draws applying to every drawn configuration (1,000 resamples per cell, all successful), repeating certification, class selection and the mixed model in each resample with the A7 weights unchanged (`paper/pilot/bootstrap_two_way.py`). Its one-sided p is obtained by recentring the resampled estimates under $H_0 : \beta_{\mathrm{DSA}} = 0$ (the share of resamples with $\beta^* - \hat{\beta} \leq \hat{\beta}$); the share of resamples with $\beta_{\mathrm{DSA}} \geq 0$ and the percentile interval are reported descriptively. This check complements the registered mixed-effects test and does not replace it. The construct checks were registered as hypotheses but not operationalised; the statistics reported in §6.3 (correlations over (configuration, class) cells, a sums-of-squares decomposition and a seed component from the repeated runs) were fixed after the pilot data were in, so they are post hoc operationalisations of registered hypotheses (`paper/pilot/construct_validity.py`). Runs that end without a decision count as errors. Decisions recovered from text rather than a tool call are included and flagged, and excluded in a robustness check. For multi-run configurations the majority-vote variant of every measure is reported alongside the per-run variant.

5.8 Deviations recorded during the pilot

A smoke test (Haiku 5.5, 462 runs; gpt-oss 20B, 136 runs) preceded the pilot and was used only to debug the harness and the policy; its data are excluded. It exposed one generator defect (the onboarding file lacked a document identifying the beneficial owners) and ambiguities in rule R2, which was clarified in four respects before generator v1.0.0 was frozen. The specification had left a policy-ambiguity audit as an open question. It was carried out after interim inspection of the first calibration data and identified two contested normative answers (amendment A1, post hoc): removing the register excerpt in the freshness class, where the policy defines when an excerpt is current but not that a file must contain one, and an inverted notarial deed in the discrepancy class that contradicts the shareholder excerpt, where reporting the discrepancy is defensible. Primary analyses keep these arms; a sensitivity analysis excludes them. Late in the pilot, before the Opus held-out run and while the last local runs were arriving, the author reviewed a sample of the oracles' answers (amendment A13, post hoc); his verdicts define a reviewed answer key, which is reported as a further sensitivity analysis (§6.1); one verdict was withdrawn the next day, after a later case, and the key was recomputed. Two amendments concern the analysis and were written after interim results were known, which the log discloses: A7 replaced an arbitrary tie-break by the expected-error rule of §5.7, after it was found that every informative comparison of the delegation test at the time involved a tie at the boundary of the delegated set, and A11 fixed the remaining baselines, measures and the mixed model before the Bielik, Gemma, Qwen and Opus held-out data existed, with interim data of the other configurations known. The judge's rubric was frozen (A2) when aggregate results for the API models, Basal and d1 were already known, before any decision had been read for that purpose, and revised once after a 20-decision trial (A3). Two code audits (A8, A12) recomputed every reported number from the raw data with separately written code; the corrections they produced are logged, and none changed a count in the delegation test. The analysis plan (`paper/pilot/preregistration.md`) is dated 25 October 2026 and was frozen in the author's own repository before the first calibration run, according to the author's records; every later change was recorded as a dated amendment. The plan was not deposited with an independent registry. The repository was created on GitHub on 25 October 2026, and the earliest push that GitHub's event log still retains (28 October 2026) already contained the plan and the first pilot data, which attests that the plan existed by that date but not that it preceded the first runs; the public release of 1 November 2026 does not by itself attest its date.

Four further amendments changed the set of configurations or their samples; each was written before the data it concerns existed.

A4 equalised sample sizes: planned sizes had differed for budget reasons, and with 15 base episodes per class Opus 5.5 could not be certified even with no errors. Opus and Sonnet were extended with the same seeds, and a common-core analysis on the shared 60 / 30 / 60 episodes and run 0 is reported with the same prominence as the registered analysis. A5 and A6 added OpenAI configurations, because every closed configuration until then came from the developer of the AI system used to prepare this work (see Statements); the most capable pair (Claude Fable 5.1, GPT-6 Astra) was not run, for cost. A9 dropped gpt-oss 20B, which duplicated the gpt-oss family; the question it would have answered, whether basis failures change with size within a family, is covered by the two three-size series.

6 RESULTS

Three decision classes are tested (§5.1). We call them the *discrepancy class* (c3 in the tables: triage of a disagreement between the beneficial-ownership register and other evidence), the *freshness class* (c5: whether a customer file must be reopened, refreshed or left alone) and the *onboarding class* (c6: an assembled onboarding file that the agent may only escalate or complete). A *configuration* is a model with its protocol (§5.3). The *common core* is the set of 60 calibration, 30 held-out and 60 shifted base episodes per class that every configuration shares (amendment A4); the analysis as registered uses all data, and we report both. Results are scored under three answer keys: the *registered key* (the oracles' answers as fixed in the analysis plan), the registered key *without the two contested arms* (amendment A1) and the *reviewed key* (the registered key with the author's verdicts applied, A13).

6.1 Delegation on interventions versus track records (the delegation test, H3)

The registered test favours DSA. For each configuration and coverage level $k \in \{1, 2\}$, each rule delegates the k classes with the lowest certified upper bound, and we compare the outcome error on the shifted episodes of the delegated classes (§5.6). With 14 configurations there are 28 such comparisons; a comparison is informative only when the two rules delegate different classes. In the registered analysis on all data, the rules differ in 10 comparisons. DSA has the lower shifted error in 9 and the higher in none (one comparison is equal), and the mixed-effects model gives $\beta_{\mathrm{DSA}} = -0.36$ (standard error, SE, 0.11; odds ratio 0.70; one-sided $p < 0.001$), where β_{DSA} is the model's coefficient for the DSA rule and a negative value favours DSA. The common core, reported with equal prominence, agrees: DSA is lower in 9 of 11 differing comparisons and higher in none (sign test $p = 0.002$), with $\beta_{\mathrm{DSA}} = -0.38$ (SE 0.11; $p < 0.001$). Because the model treats a shifted episode that enters both rules and both values of k as independent observations, we also resampled configurations with replacement (1,000 resamples): the bootstrap standard errors are 0.12 on all data and 0.14 in the common core, the one-sided p-values from a normal approximation with the bootstrap standard error 0.001 and 0.003, and no resample gives $\beta_{\mathrm{DSA}} \geq 0$ (a descriptive share, not a test). The two-way bootstrap of §5.7, which also resamples episodes, gives bootstrap standard errors of 0.22 on all data and 0.21 in the common core, one-sided recentred p-values of 0.052 and 0.033, 95% percentile intervals of [-0.80, +0.04] and [-0.79, +0.01], and 5% and 3% of resamples with $\beta_{\mathrm{DSA}} \geq 0$ (Table 4): once the sampling of episodes enters the uncertainty, the registered advantage is significant at the 0.05 level in the common core only, and not on all data. By the registered criterion, the delegation test passes; §8 reports the result under all 60 logged readings of the policy and under 30 further readings that drop the policy's own independence condition.

The advantage rests on one contested arm. A policy audit carried out during the pilot, after interim inspection of the first calibration data, had flagged two intervention arms whose correct answers are contested (amendment A1, post hoc): the removal of the register excerpt in the freshness class, where the policy does not say that a file must contain one, and an inverted notarial deed in the discrepancy class. Excluding both leaves 7 differing comparisons, 4 : 3, in both data sets ($\beta_{\mathrm{DSA}} = -0.09$, $p = 0.19$ in the common core; -0.08, $p = 0.20$ on all data). The first arm carries the result: dropping the removal arm in every class has the same effect (common core, $p = 0.38$ in the leave-one-out check), dropping the inversion arm does not ($p = 0.004$), and in the two reviewed cases of the inverted deed the author's blind answers equalled the oracles' (A13). Rescoring with the reviewed key, described next, gives 3 : 1 in 8 differing comparisons on all data ($\beta_{\mathrm{DSA}} = -0.07$, $p = 0.28$) and 3 : 1 in 9 in the common core (-0.08, $p = 0.28$).

The author's review of the key. Late in the pilot the author reviewed 21 of 30 sampled cases, drawn where models disagreed with the oracles, so the result is not an error rate of the oracles (A13). He answered them in three steps: 11 blind, on paper; 9 blind after the four open readings had been discussed; and one after the second batch's answers had been shown to him. His answers equalled the oracles' admissible sets in 14 of the 21 (6 of 11 in the first batch; 7 of 9 in the second, where the two differences were a date copied wrongly onto his own board and a case in which he listed three actions where the oracles admit one; and the last case), and he reports that elements such as the decision date came into view only after several cases, so later answers are not independent of earlier ones. He first judged the oracles wrong in one case: exception R2(b) requires a source independent of the register's controlling authority, the documents name issuers but never their controlling authorities, and both oracles decide independence from a field the agents do not see, so he read the exception as never established. He withdrew this verdict the next day, after answering blind a case of the same structure in the oracles' sense, on the ground that a register and a court are two independent bodies even though no document says who controls them. The reviewed key therefore keeps the oracles' reading of the exception; the legal text that R2(b) paraphrases has no independence condition at all (§8). He judged both readings defensible in three cases, which turn on two questions the policy leaves open: whether filing annual accounts is a "change" that requires reopening a file (R3), and whether a file must contain a register excerpt at all (R3, R5). The two oracles were written from one specification and share these readings, so their agreement did not test them. Together the two readings change the correct answer for 44 task variants in the common core. The clearest case is the freshness class. In the common core the track record certified it for GPT-6 Luna, GPT-6.1 Terra, GPT-6.1 Sol and Qwen 3.8 27B, and DSA did not. Under the registered key these configurations then erred on 17–27% of the shifted freshness episodes, which made the

Table 4: The delegation test under the registered key, without the contested arms, and under the reviewed key. "Comparisons": (configuration, k) comparisons in which DSA has the lower : higher shifted error, out of those in which the two rules differ. β_{DSA} with, in parentheses, the one-sided p from the mixed-effects model, the one-sided p from a normal approximation with the standard error of a bootstrap over configurations (1,000 resamples; `paper/pilot/bootstrap_h3.md`), and the one-sided recentred p from a two-way bootstrap over configurations and episodes (1,000 resamples; post hoc, §5.7; `paper/pilot/bootstrap_two_way.md`).

Analysis	All data: comparisons	All data: β_{DSA} (p)	Common core: comparisons	Common core: β_{DSA} (p)
Registered key	9 : 0 of 10	−0.36 (< 0.001; 0.001; 0.052)	9 : 0 of 11	−0.38 (< 0.001; 0.003; 0.033)
Contested arms excluded (A1)	4 : 3 of 7	−0.08 (0.20; 0.27; 0.29)	4 : 3 of 7	−0.09 (0.19; 0.26; 0.26)
Reviewed key (A13)	3 : 1 of 8	−0.07 (0.28; 0.15; 0.18)	3 : 1 of 9	−0.08 (0.28; 0.16; 0.18)

track record look unsafe. Under the reviewed key they erred on 0–2%: almost every one of these errors was reopening a file after an annual-accounts filing that the shifted split had added as a decoy.

What holds under every key. Under each key, the LLM judge passes most of the correct-outcome decisions that fail an interventional test: 63–70% on all data and 59–68% in the common core, pooled over the configurations it assessed (the judge test; §6.2). It also flags 17–18% of the decisions that pass every intervention, so the two methods disagree in both directions, and the pilot has no third standard to say which is right. As delegation rules, the two judge-based rules had a higher shifted error than DSA in most comparisons in which they differed, under every key. In about two thirds of these comparisons, however, DSA delegated the onboarding class and the judge-based rule did not, and in that class the judge mostly reads the "current register excerpt" of R4 as an excerpt of the beneficial-ownership register, unlike the oracles and the agents (interim observation 2 in the registration log). These comparisons are descriptive. In the common core, the readiness gate, a two-arm rule defined for the API configurations, had the lower shifted error in 2 of 6 differing comparisons under the registered key (the other 4 equal), in both of the 2 differing comparisons without the contested arms, and split 1 : 1 under the reviewed key (§6.2).

What each rule certified. Under the registered key the DSA rule certified only the onboarding class, in 7 of 14 configurations (Basal, Opus, Sonnet, Bielik, Qwen, Sol and d1), from 60 calibration episodes per class in both data sets (§7.1 gives our reading of the test); in that class escalation is always an admissible answer. In the common core the track record certified 16 classes, with a shifted error of 5.7% in them. Without the contested arms DSA certified 11 classes with a shifted error of 5.8%, about the same as the track record's; under the reviewed key DSA certified 10 classes with one shifted error in 600 episodes and the track record 16 with three in 960 (`paper/pilot/construct_validity.md`, which also gives the all-data figures). Eight of the eleven differing comparisons in the common core involve a tie at the boundary of the delegated set, resolved by the expected-error rule of A7. Restricted to the discrepancy and freshness classes, in which delegation would mean autonomy over a decision, the comparison at $k = 1$ gives 6 : 0, 2 : 2 and 1 : 1 differing comparisons under the three keys (with two further equal comparisons under the reviewed key). Two further caveats apply. The reviewed key rests on one reviewer, who saw exploratory results before giving three of his four verdicts and who withdrew one of them after a later case, knowing the direction of the change (A13). The configuration bootstrap of Table 4 (first proposed at the A12 audit, when an interim run on 11 configurations had given a standard error of 0.149 against the model's 0.126) leaves every reading unchanged; the sign test's p-values are descriptive (§5.7).

6.2 Reading the trace versus intervening on the evidence (the judge test, H1′, and the declared-grounds test, H2)

The judge passes most wrong-basis decisions. The judge test asks what share of the correct-outcome decisions that fail an interventional basis test a procedure audit passes. The judge (Claude Haiku 5.5, rubric v2 after one revision, amendment A3) read the factual decision, its explanation and its declared grounds for every agent with tools and for the one-pass model; decision models return no trace to read. Pooled over the eleven judged configurations, the audit passed 322 of the 474 correct decisions that fail a basis test in the common core (67.9%) under the registered key, 169 of 288 (58.7%) without the contested arms and 205 of 337 (60.8%) under the reviewed key; on all data, 70.3%, 63.3% and 64.7%. It also flagged 17–18% of the correct decisions that pass every intervention: 28.5% in the onboarding class against 6–12% in the other two, because there it reads the "current register excerpt" of R4 as an excerpt of the beneficial-ownership register (§6.1). Per configuration, under the registered key, the judge passed 74–100% of the basis failures of the six API agents (all 25 for GPT-6.1 Sol, 18 of 19 for Claude Opus 5.5) and 37–56% of those of three local agents with tools (Gemma 4 26B, gpt-oss 120B, Mistral Small 3.2) and the one-pass model, whose explanations are shorter and whose failures are more often outright non-reactions (§6.3); the fourth local agent, Qwen 3.8 27B, which rarely fails by non-reaction, sits with the API agents at 85%. The main judge also judged Claude Haiku 5.5's own decisions (35 of the 474 basis failures, 26 of them passed); excluding them gives 296 of 439 (67.4%).

A stronger judge does no better. The main alternative explanation, that Haiku is weaker than several of the models it judged, was tested before any verdict of the second judge was produced (amendment A10). GPT-6.1 Sol judged 199 correct-outcome decisions of five configurations (Claude Haiku, Sonnet and Opus 5.5, GPT-6 Luna and GPT-6.1 Terra; its own decisions were excluded), drawn in strata with and without an interventional basis failure, with the same rubric. Its Youden's J (sensitivity minus false-positive

rate) was −0.07 against +0.07 for Haiku, difference −0.14 with a 95% bootstrap interval of [−0.24, −0.04]; it flagged none of the 40 basis failures of the OpenAI agents, and its mean grounds score was 4.88 for decisions with a basis failure against 4.77 without. The two judges agreed on 87% of verdicts (Cohen's $\kappa = 0.31$). By the reading fixed in advance ($J < 0.15$; fixed before any verdict of the second judge, after the first judge's results were known, amendment A10; on the final pool Haiku's own J is about 0.15), the "convincing explanations" account is supported: the explanations of decisions that do not depend on the decisive evidence read as well as those that do, to two judges from two developers. Relabelling the same 199 decisions under the other two keys changes little: without the contested arms 38 of them count as basis failures instead of 99, and Sol's J is −0.03 against +0.10 for Haiku; under the reviewed key 52 count, and Sol's J is −0.06 against +0.07; the difference is −0.13 in both cases, with bootstrap intervals [−0.25, −0.03] and [−0.24, −0.03] (`runs/judge_check/result_keys.txt`).

Declared grounds carry a signal, but a weak one. The declared-grounds test compares the interventions with a static check of what the agent itself named as decisive: the judge's verdict on whether the documents the agent declared as decisive are those that settle the episode under the policy. As registered, the interventions are the reference, so what follows is the sensitivity and the false-positive rate of the static check against them, not a symmetric comparison of two detectors. This check flagged 65% of the interventional basis failures and 35% of the decisions that pass every intervention (common core, registered key; 56% and 41% without contested arms; 59% and 39% under the reviewed key). Table 6 in Appendix A gives the confusion matrices. Taken as a detector of what the interventions find, it separates better than the judge's basis verdict (sensitivity 0.65 against 0.32, false-positive rate 0.35 against 0.17), but it flags a third of the decisions whose basis the interventions confirm, mostly for naming too many or too few documents. A mechanical version of the check, which compares the declared decisive documents with the document the generator defines as decisive, does not separate at all: any difference between the two sets flags 62% of the basis failures and 84% of the decisions that pass every intervention, and a decisive document left undeclared flags 13% and 26% (`paper/pilot/declared_grounds_check.md`). Agents name the register entry together with the evidence that confirms or contradicts it, and the generator's list names the settling document alone, so what separates is the judge's reading of the declaration against the policy, not a match with a fixed list. The test is met in the weak sense that interventions and declared grounds disagree in both directions and the interventions find failures that the declared grounds do not show.

As delegation rules. The judged-process rule certified five classes in the common core under the registered key and the same five under the reviewed key (the discrepancy class for Opus and Sol, the freshness class for Luna, Sol and Terra); the judged-grounds rule, with its threshold chosen on calibration (3.25 on a five-point scale), granted every class to every API agent and to Gemma 4 26B and Qwen 3.8 27B, including the onboarding class for GPT-6.1 Terra with a calibration outcome error of 46.7%, and withheld every class but one from the other local models: mean grounds scores separate verbose from terse configurations, not reliable from unreliable classes (interim observation 3). At matched coverage, DSA had a lower shifted error than the judged-process rule in 11 of 13 differing comparisons under the registered key (none higher, two equal), 7 of 10 (one higher) without contested arms and 6 of 11 (none higher, five equal) under the reviewed key, and than the judged-grounds rule in 13 of 14, 9 of 11 (one higher) and 8 of 12 (none higher); as §6.1 notes, about two thirds of these wins are comparisons in which DSA delegates the onboarding class and the judge-based rule does not. The readiness gate, defined only for the API configurations, had the lower shifted error in 2 of 6 differing comparisons under the registered key, with the other 4 equal, and split 1 : 1 of 6 under the reviewed key. With the contested intervention arms excluded, the readiness gate and DSA differed in two matched-coverage comparisons, and the readiness gate had the lower shifted outcome error in both. These comparisons are descriptive and too few to establish an advantage for either rule; the pilot gives no evidence that the full intervention family delegates better than this two-arm rule. Two further baselines could not certify at this sample size, which the design implies rather than the data: the B-conf threshold rule cannot certify any threshold because its Bonferroni correction over candidate thresholds requires 74–78 error-free decisions above a threshold, more than the 60 calibration decisions a class has (A11.2), and a simulated reviewer who errs with probability 0.05 grants even a perfect agent with probability at most 0.19 (A12). The decision models' confidence separates correct from incorrect outcomes (AUROC 0.55–0.78) about as well as it separates right from wrong basis among correct outcomes (0.56–0.84 under the registered key, 0.56–0.81 under the reviewed key), with Basal 2.0-max the highest on the basis measure. The reported confidence is overconfident for Basal 2.0-max and Liquid AI d1-3B (expected calibration error against outcome correctness 0.16 and 0.20; mean confidence 0.84 and 0.66 against accuracy 0.68 and 0.57) and closer to calibrated for GPT-6 Luna on the Decisions API (0.09; 0.13 on the probability of the chosen action, which its confidence field understates). As a deferral rule, sending the half of the decisions with the lowest confidence to a human lowers the unjustified rate among the rest from 61% to 34% for Basal and from 66% to 52% for Luna, catching 72% and 60% of the unjustified decisions; for d1 the rule helps at a quarter deferred (67% to 56%) and gets worse at half (63%), because its most confident decisions are mostly wrong (Table 5 in Appendix A).

6.3 Construct checks (CV1, CV2) and the known-groups check

In this pilot DSA is largely, not wholly, reducible to outcome accuracy (CV1). Across the 42 (configuration, class) cells of the common core, the share of correct outcomes that fail a basis test rises with the outcome error: Pearson $r = 0.88$ (Spearman $\rho = 0.79$) under the registered key, 0.94 (0.94) without the contested arms and 0.93 (0.91) under the reviewed key; a linear regression on ℓ_{out} leaves 23%, 11% and 14% of the variance unexplained, and 17–22% across configurations with classes pooled (`paper/pilot/construct_validity.md`). The expectation stated in the analysis plan, that DSA is not reducible to outcome accuracy is therefore only weakly met: configurations that err more also fail more basis tests when they are right, and the independent part is between a tenth and a quarter of the variance. The residual sits

where the construct predicts. Agents with no outcome error in a class still fail basis tests in 8–17% of discrepancy episodes (Claude Haiku, Opus and Sonnet 5.5, GPT-6.1 Sol) and in 33–40% of freshness episodes (GPT-6 Luna, GPT-6.1 Terra and Sol, Qwen 3.8 27B; Sonnet, with an outcome error of 3%, in 43%), and for these agents a large share of the basis failures, for Sonnet, Opus, Sol, Luna and Qwen most of them, lie on the contested arms (counting correct-outcome decisions with a basis failure: 26 of 35 for Sonnet, 14 of 19 for Opus, 20 of 25 for Sol, 24 of 44 for Luna, 22 of 46 for Terra, 14 of 35 for Haiku and 27 of 39 for Qwen 3.8 27B), whereas for the local and decision models few do (7 of 70 for gpt-oss 120B, 6 of 53 for Basal) and the failures are mostly non-reactions: Liquid d1 kept its factual action in 100% of the discrepancy-class interventions that made it inadmissible, Basal 2.0-max in 81% and Mistral Small 3.2 in 51%, against 0–7% for the API agents. Whether the frontier models' residual is dependence on evidence or disagreement with the key is what §6.1 leaves open.

Where the variance lies (CV2). Decomposing the unit-level basis loss of the 2,520 core calibration units by sums of squares, 15% lies between configurations, 20% between classes, 13% in their interaction and 52% within cells (registered key; 27%, 10%, 22% and 42% without contested arms; 26%, 12%, 20% and 43% under the reviewed key). Most of the variance is between episodes of one configuration and class, which is where the certification bound operates and where a sample of 60 is small. The seed component can be measured only for Claude Haiku 5.5, which ran three times: its runs disagree on the basis loss in 22% of episodes, and 29% of the variance of its basis loss lies between runs of the same episode (54% without contested arms, 55% under the reviewed key). For the thirteen configurations that ran once, sampling noise and dependence on evidence cannot be separated (§8).

Known groups. The five scripted agents largely behaved as predicted in expectations written down before the run (file `known_groups.md`). The oracle agent is clean under every measure. The memorising agent, which replays stored answers for episodes it has seen, has zero calibration outcome error in all three classes, so the track-record rule delegates all of them, while DSA withholds the discrepancy and freshness classes (basis loss 80% and 100%) and the agent's held-out and shifted errors in those classes are 43–60%: a perfect track record with no dependence on evidence, the case that motivates the paper (§1.2). The always-escalate agent is clean in the onboarding class and fails the other two under both rules. The independence-blind agent errs only where independence decides (discrepancy class: outcome error 10%, basis loss 25%, shifted error 25%; the other classes clean). The cue-following agent flips under placebo and sham arms in 28% of discrepancy and 16% of freshness arms and never in the onboarding class. One expectation was not met: the copy arm did not reveal the independence-blind agent (copy flip 0%), because adding a dependent copy of the decisive document does not change the oracles' answer in these episodes; the deficit shows only through the add-source arms and the shifted split. The check validates the code and the measures on deficits we planted; it does not bear on the delegation test, and in the onboarding class both rules certify every planted agent, because escalation is always admissible there.

7 DISCUSSION

7.1 What the results say about delegation

The registered question is not settled. We asked whether an agent's reaction to changes in the evidence is a better basis for delegating a class of decisions than its track record. The registered test passes, and we do not read it as support (§6.1): the advantage rests on one intervention arm, the removal of a register excerpt the policy does not require, whose correct answer a policy audit flagged during the pilot (the second arm the audit flagged, an inverted deed, does not carry it; §6.1), it disappears when the flagged arms are excluded or when the author's reviewed key is used, and in the exploratory multiverse of 60 combinations of four logged policy readings (§8) it reaches a nominal one-sided $p < 0.05$ in ten of the 60 keys, without correction for multiplicity, each of which includes a reading stricter than the legal texts the policy paraphrases; one of the ten is the key that follows the policy's own wording with reporting admitted ($p = 0.011$ on 7 comparisons; §8). Under the 30 further keys of the sensitivity analysis that drops the policy's own independence condition, which the legal text does not contain, it reaches $p < 0.05$ under none (§8). The sensitivity analyses are post hoc, and the reviewed key rests on one reviewer. The registered estimate itself is less certain than the model reports: when the sampling of episodes enters the uncertainty (the two-way bootstrap of §5.7), its one-sided p is 0.052 on all data and 0.033 in the common core, and the 95% intervals reach +0.04 and +0.01. In sum, the pilot cannot tell the two rules apart, and the registered result would have been reported as a success had the answer key not been examined. Among the API configurations, a readiness gate built from only the null arm and the removal of a sole source was never worse than the full intervention family at matched coverage under the registered key and without the contested arms, and split 1 : 1 with it under the reviewed key (§6.2); the pilot gives no evidence that the composed arms add delegation value.

The key fed both sides of the test. The certification step and the criterion it was judged by were scored against the same oracles. In the freshness class this produced an advantage that was itself right for the wrong reason (amendment A11, note 6): DSA withheld the class because the agents answered "no action" on an arm whose correct answer the policy leaves open (a file without a register excerpt), and the track record's grants of the same class then looked unsafe because of a different convention that exists only in the oracle code (a filing of annual accounts is not a "change"). A delegation rule inherits every ambiguity in the normative map, and so does any criterion scored against that map. For a delegator the consequence is practical: before two rules are compared, the answer key needs the governance that a model gets. That means an owner, documented readings of every open-textured term, admissible sets rather than single answers where the policy is silent, blind review of a disagreement sample by more than one person, and a sensitivity analysis over the logged readings. In our pilot this review would have cost a fraction of the runs, and it would have been cheaper still before them.

What the pilot does show about agents. Right outcomes on a wrong basis occur in every configuration, but they mean different things at the two ends of the range. For the frontier agents between two fifths and four fifths of the basis failures lie on the contested

arms, so their extent is a matter of the key (§6.3). For the local and decision models the failures are mostly non-reactions to the decisive evidence (§6.3), the deficit the known-groups check planted in the memorising agent and the instrument detected. Within the two three-size series, the basis loss ℓ_{basis} falls with size in the discrepancy class for the GPT-6 line (32%, 30%, 8%) and in the freshness and onboarding classes for the Claude line (53%, 45%, 42%; 7%, 0%, 0%; common core, calibration split, `paper/pilot/results_core.md`), and not in the other three series–class pairs; with 60 episodes per class these differences are descriptive. Basal 2.0-max's reported probabilities, although overconfident (Appendix A), predict its own basis failures (AUROC 0.84), which suggests that decision models carry information about the basis of their answers that a confidence threshold could use, but the threshold rule cannot certify anything at this sample size.

Certification at sixty episodes. With $\alpha = 0.10$ and one failure tolerated, a class is certified only when it is almost clean. Under the registered key DSA certified only the onboarding class, in which escalation is always admissible, so its grants delegated little autonomy; the track record, which counts fewer failures, certified more classes, and more of them went wrong after the shift under the registered key, and none of this survives the reviewed key. A reader should not take a DSA grant in this pilot as a recommendation to automate anything. The sample sizes the design needs for a confirmatory test are those of the second phase.

7.2 Why reading explanations does not certify a basis

The judge test gives the pilot's most stable result. Two judges from two developers, one of them a flagship model, read the decision, the explanation and the declared grounds, and both pass most of the correct decisions that fail an interventional test; their Youden's J is near zero, and they rate the explanations of wrong-basis decisions as well as the others (§6.2). Three things follow.

First, an explanation is a report of one execution, and whether a decision depends on a piece of evidence is a relational property that cannot be read from one execution (Pandey et al. 2026a). A judge can check that the stated reasons are consistent with the policy, which is what it did, and that is a different property. Stated reasons that read well and are unfaithful are a known failure mode (Turpin et al. 2023; Lanham et al. 2023); the pilot adds that a second model reading the same text passes most of the decisions that fail an interventional test. Second, a judge brings its own reading of the policy. In the onboarding class the Haiku judge read "current register excerpt" as the beneficial-ownership register, unlike the oracles and every agent, and that one reading produced about two thirds of the comparisons the judge-based rules lost (§6.1). A judge-gated autonomy rule, as proposed in delegation frameworks with behavioural metrics (Tomašev et al. 2026) or implemented in readiness qualification with an LLM-judged process defect (Chatrath et al. 2026), therefore adds a second answer key rather than an independent check. Third, the signal that does exist in the trace is structural. The judge's check of the documents the agent named as decisive separated wrong-basis decisions from the rest better than its free-text verdict, and a mechanical comparison with a fixed list of decisive documents did not separate at all (§6.2): what carries the signal is a reading of the declaration against the policy. Declared grounds, as the typed decision records reviewed in §2 propose, are the part of a trace worth keeping, but checking them is itself an act of interpretation.

The limits of this result are the limits of the pilot. There is no third standard: the judge also flags about a sixth of the decisions that pass every intervention, and where the judge and the oracles read a rule differently, as in the onboarding class, the judge may be the one that is right. The judges used one rubric, revised once (A3), and the test of a stronger judge, although its result holds under all three keys, rests on one sample of 199 decisions from five configurations. What we can say is that in this pilot reading the explanation caught few of the basis failures the interventions found, that the pilot offers no substitute for intervening, and that an evaluator who wants that check has to intervene on the evidence or verify the declared grounds against the record, with an answer key that has been reviewed for the purpose.

7.3 What a delegated decision costs

The price of model inference keeps falling, yet the total cost of running agents can still grow faster than the price falls. Cheaper decisions invite more of them, more autonomous steps per task and more tasks handed to agents, a rebound effect familiar from energy economics (Jevons 1865, p. 103). For an organisation the relevant quantity is therefore not the price of a million tokens but the cost of a decision that can safely be left to an agent.

The cost function of §4.7 makes this explicit. Inference (c_c^{inf}) is one term among several, and usually the smallest. A delegated class also carries the expected cost of the errors that the certificate tolerates ($\varphi_c L_c R_c$); a withheld class carries the cost of human review ($e_c + h_c L_c$), measured in minutes of a specialist's time rather than in tokens; and every certified class carries the cost of producing the evidence (C_c/N).

The pilot gives a sense of the magnitudes, from the token counts in the run logs (`paper/pilot/token_costs.md`). A factual decision by an agent with tools took three turns and 4,000–8,000 input-side tokens for the API models (Gemma 4 26B, with six turns, 14,000), and a decision model took one call of 1,400–2,600 tokens. Certifying one class from 60 episodes, averaging eight or nine arms per episode, took 7.5–9 times the tokens of the 60 factual decisions. At the list prices recorded in the specification (6 October 2026, checked against the pricing page on 31 October 2026), a factual decision cost between USD 0.0001 (GPT-6 Luna on the Decisions API) and USD 0.035 (Claude Opus 5.5), and certifying one class between USD 0.05 (the Decisions API; USD 0.22 for GPT-6 Luna with tools) and USD 13.69 (Claude Opus 5.5), six to seven times the cost of the factual decisions in the same calibration set for the Claude configurations and eight to nine times for the OpenAI ones (Claude rates as recorded on 6 October 2026, with Sonnet's cache-read rate corrected on 31 October; OpenAI rates as recorded in the run code on 26 October 2026). The local models and the two local decision models ran on the author's hardware and are not priced. The providers' usage exports for the whole pilot (25–31 October 2026) show USD 116.21 billed by Anthropic (Haiku 17.78 including the judge and the smoke test, Sonnet 30.16, Opus 68.27) and USD 23.63 by OpenAI (the three agents, the Decisions API and the second judge); the token counts in

the run logs agree with the billed counts to within 0.1% in aggregate, and the logged tokens at the rates above come to within USD 0.04 of the billed total (`paper/pilot/billing_check.md`), so the figures in this section are the billed rates, not a list-price estimate. The cost ratio is below the token ratio because the intervention arms re-read the unchanged documents from the prompt cache, billed at a tenth of the input price for Haiku and Luna and a twentieth for Sonnet, Opus, Terra and Sol at the recorded rates: for the Claude configurations cache reads were 58–81% of the input-side tokens of a factual decision and 86–94% of those of an intervention arm. This cost is paid once per configuration and class and then spread over every decision the certificate covers, plus the cost of sequential monitoring (§4.6). Against a human review of the same file, which takes minutes, it is small.

Two consequences follow. First, the cheapest model per token is not necessarily the cheapest per delegated decision: a configuration that cannot be certified in a class sends that class to human review, and a configuration certified on its track record alone can cost more through errors after a shift. Second, verification has to be priced into deployment from the start, as part of the cost of a decision, not added as an audit afterwards. A delegation rule that certifies the basis of decisions is, in this sense, also a cost-control mechanism: it says where autonomy pays for itself and where it does not.

7.4 Applying the instrument where the agent is a product

A reader whose institution has one AI tool, an office assistant such as Microsoft 365 Copilot, will ask how any of this applies. The instrument needs three things from a configuration, none of which is a particular model: it must be callable programmatically, since certifying one class took about five hundred runs in the pilot and monitoring continues in production; its inputs must be controllable, since every intervention replaces one document in the file and the agent must not see the other variants of the same case; and it must return a decision with declared grounds. An assistant used by hand in a chat window meets none of these. An agent built in the institution's own tooling, with the policy as its instructions and a tool that fetches the documents of a case, exposed through that tooling's programmatic interface, meets all three; at the time of writing (October 2026) the Microsoft stack offers this through Copilot Studio agents called from the institution's own code (Microsoft 2026), and the GitHub Copilot SDK puts frontier models from several developers behind custom tools (GitHub 2026), the form the pilot's agents took directly on the developers' APIs. What is certified is then the artefact that would be deployed. Two consequences follow from the results above. A product that selects and updates its models silently changes the configuration under the certificate, so certificates are short-lived and sequential monitoring (§4.6) becomes the main instrument. And the convenient reviewer, the same assistant asked to check its own decisions, is the judge of §6.2, which passed most of the wrong-basis decisions it saw. The oracles remain the hard part: an institution has to write down its own readings of its own policy before any agent runs, which is the lesson of §7.1.

8 LIMITATIONS AND THREATS TO VALIDITY

What DSA measures depends on the intervention family and on the normative map. A class is certified relative to the interventions we run and the answers our oracles assign to them. Two independent oracles guard against implementation errors, not against a wrong reading of the policy, and both were written, like the generator, by the AI assistant used throughout (§5.2); versions of a program written independently still fail on the same inputs (Knight and Leveson 1986). The ambiguity log (`paper/pilot/policy_ambiguities.md`) lists twelve places where the policy text admits more than one reading; six affect the pilot's results. Two were found by a policy audit during the pilot, after interim inspection of the first calibration data (a file without a register excerpt, and an inverted deed that contradicts the excerpt; amendment A1). Two were exposed by the LLM judge: whether review intervals (R3) bind a register-discrepancy decision, and whether the "current register excerpt" of R4 means the business register or the beneficial-ownership register. Two more were raised when the author reviewed a sample of the oracles' answers (amendment A13): whether the documents show that two sources are independent, which exception R2(b) requires (an objection the author withdrew after a later case; §6.1), and whether filing annual accounts counts as a change. The review also judged both readings of the excerpt rule defensible. Under the reviewed key the advantage of DSA in the delegation test disappears (§6.1). We keep primary analyses as registered and report sensitivity analyses. The general lesson is that interventional ground truth is only as good as the policy it encodes, as test labels are only as good as their annotation (Northcutt et al. 2021), so in deployment the normative map needs the same review as the policy itself.

The policy is stricter than the texts it paraphrases, and that is where the readings diverge. After the review we checked the disputed readings against the public texts of §5.1; the check, with quotations, is in the ambiguity log. The FATF Recommendations and the EBA guidelines settle none of them. They require "reliable, independent" sources and leave it to each firm to "set out in their policies and procedures which information and data they will treat as reliable and independent" (EBA guideline 4.27), where independence means independence from the customer, not between sources; and they tie re-review to "a change in the customer's circumstances" (4.78) without listing filings. The AMLR, applicable from 10 July 2027, is more specific, and against it the policy is stricter in two of the contested places. Its discrepancy derogation applies when the owners "are known to the obliged entity from another reliable source" (Art. 24(2)(b)); the condition that the source be independent of the register's controlling authority, which the author's review first found impossible to establish from the documents and then accepted as the ordinary reading, is R2(b)'s own. The derogation is also a permission ("may refrain from reporting … and may instead request additional information", Art. 24(2); recital 54 adds that it "should be possible" to let the customer report minor, typographical differences), so reporting stays admissible where it applies, and the oracles' single answer narrows it. Article 26 sets the same review intervals as R3 but triggers an earlier review on "a change in the relevant circumstances of a customer" (recital 69: "material changes"), so R3's "any change recorded in an official filing" goes

beyond the text. "Valid proof of registration or a recently issued excerpt of the register" is required only when entering a new relationship (Art. 23(4)); no text requires one in the file of an existing customer or sets its age, so the 30-day window and the excerpt requirement are the institution's choices. R4 follows Art. 76(5), which requires "meaningful human intervention" in decisions to enter, refuse or maintain a relationship taken with AI systems. All 60 keys of the exploratory multiverse (a multiverse analysis in the sense of Steegen et al. 2016; the 60 keys differ in at least one admissible set) are readings of the synthetic policy and keep its condition, in R2(b), that the other source be independent of the register's controlling authority; they vary whether that independence counts as shown (decided from a field the agents do not see) or never, whether a filing of annual accounts is a change, what a missing excerpt requires, and how the inverted deed is scored. Under them the advantage of DSA reaches a nominal one-sided $p < 0.05$ under 10 keys in the common core and 11 on all data, without correction for multiplicity. Each of these keys includes at least one reading stricter than the legal texts: a required register excerpt (seven of the ten) or annual accounts counted as a change (the other three, with reporting admitted for the inverted deed). The two keys among the 60 that take the lenient reading on the other three dimensions (independence shown, annual accounts not a change, no excerpt required), both keeping the controller condition, differ only in the inverted notarial deed of the discrepancy class, a reading the legal check does not settle (ambiguity log, entry 2): with the deed scored as the oracles do, DSA has the lower shifted error in 3 of 9 differing comparisons and the higher in 4 (common core; $\beta_{\mathrm{DSA}} = -0.01$, $p = 0.48$); with reporting also admitted for the deed, 4 of 7 and 3 (-0.09, $p = 0.19$). The key that follows the policy's own wording, which is what the agents read (annual accounts a change, excerpt optional), gives 3 : 2 of 9 (-0.01, $p = 0.48$) with the deed scored as the oracles do and 4 : 1 of 7 (-0.29, $p = 0.011$) with reporting also admitted for the deed. This nominally significant result is sensitive to the neighbouring readings and does not establish an advantage; we report it because it is the key an agent that followed R3 to the letter would be scored by, and the registered key scored agents that reopened a file after an annual-accounts filing as wrong although R3's wording told them to. Whether the derogation of Art. 24(2) is a permission, so that reporting stays admissible where it applies, is a separate question (entry 9); an exploratory check that admitted reporting in that entry's case changed no number of the delegation test, and reporting admissible wherever the derogation applies was not varied as a reading of the multiverse: every key keeps the oracles' single-action convention where the exception applies, and "reporting also admitted" concerns only the contested inverted deed. The key built from the author's blind answers alone favours the track record in direction (DSA lower in 2 of the 5 comparisons with different selections, higher in 2, equal in 1; $\beta_{\mathrm{DSA}} = +0.19$, $p = 0.95$).

Dropping the policy's independence condition. The AMLR requires only that the owners be known "from another reliable source" (Art. 24(2)(b)); no key among the 60 implements that reading, since the controller condition is the policy's own. As a separate sensitivity analysis added after a review (3 November 2026), we re-scored the discrepancy class with the condition dropped, so that exception (b) applies whenever another reliable source shows the current owners. This changes the admissible set of 1,162 of the 34,580 records in the common core and 1,491 of 45,001 on all data, and combined with the other three readings gives 30 further keys (rows 61–90 of `paper/pilot/multiverse/multiverse.md`). Under none of them does DSA reach a nominal $p < 0.05$: with the deed scored as the oracles do, p runs from 0.13 to 0.99 in the common core (0.15 to 0.96 on all data); with reporting also admitted, from 0.25 to 1.00 (0.27 to 0.99). The key that drops the controller condition and takes the lenient reading on the other three dimensions (annual accounts not a change, no excerpt required, deed as the oracles do; row 63) gives DSA the lower shifted error in 4 of the 5 comparisons in which the two rules select different classes, the higher in none and an equal error in 1 ($\beta_{\mathrm{DSA}} = -0.09$, $p = 0.14$; all data 4 of 4, -0.08, $p = 0.16$); with reporting also admitted for the deed (row 64), the lower in 2 of 4, the higher in 1 and equal in 1 (-0.03, $p = 0.35$; all data 2 of 3, higher in 1). Removing the controller-independence requirement therefore yields no statistical support for an advantage of DSA in these 30 variants; they still keep the oracles' single-action convention where the exception applies, so they test the policy's condition, not every reading of the legal text, and the pilot's registered result rests on that stricter condition. In the second phase the policy text will be aligned with the legal texts before any test is run. This is our reading of the texts for the pilot, not legal advice.

DSA could reduce to capability. If configurations that decide well also always reason well, DSA adds nothing to a track record. In the pilot the two measures do differ, but by how much depends on the answer key. Sonnet 5.5, for example, reached the correct outcome in the discrepancy class while failing an interventional test in 17% of calibration episodes under the registered and the reviewed key and in 5% without the contested arms. The known-groups test shows the same difference in a constructed case: an agent that replays stored answers has zero calibration errors against the oracles, so a track-record rule delegates every class to it, while DSA withholds the two classes in which its answers do not follow the evidence (§6.3). This checks the instrument, not whether the difference helps in delegation; that is what the delegation test tested, and the pilot does not show it (§6.1). The construct check specified in the analysis plan found the two measures strongly correlated (§6.3) and still needs more model configurations than the pilot provides.

Agents may recognise tests. Some interventions produce unusual files, such as a decisive document missing from an otherwise complete record. An agent that detects such variants could behave differently on them than in deployment. Frontier models already tell evaluation transcripts from deployment ones better than chance (Needham et al. 2025), and our design does not rule out such evaluation awareness. Placebo and sham controls detect sensitivity to irrelevant changes, but not detection of relevant ones. A deployed certifier would need interventions drawn from the same distribution as live episodes.

Noise inflates basis loss. Basis loss counts a failure if any of five intervention arms is wrong, so sampling noise adds failures even when the agent's dependence on evidence is correct. Haiku 5.5 changed its answer on 10% of identical re-runs in the freshness and onboarding classes over its three runs (13% in the first run alone, the null-flip column of Table 7), its three runs disagree on the basis loss in 22% of episodes, and 29% of the variance of its basis loss lies

between runs of the same episode (§6.3). Majority voting over its three runs reduced ℓ_{WT} in the freshness class from 53.3% over the three runs (55.0% in the first run, Table 7) to 47.0% (basis failures from 49.7% to 44.0%), which shows that the noise is real but does not explain the effect. Local models ran once with a fixed seed, so their noise is not measured.

The baselines are only as strong as we made them. The judged-process baseline uses Claude Haiku 5.5 for budget reasons. A check with a stronger judge from another developer, GPT-6.1 Sol, on 199 stratified decisions found no better discrimination (Youden's J of -0.07 against $+0.07$ for Haiku; amendment A10), so a weak judge does not explain the result, but two judges and one rubric do not exhaust the design space. We revised the judge's rubric once, after a 20-decision trial, to stop it from applying rules outside the task (amendment A3). We disclose this and note that the revision favours the baseline. Approval and correction baselines are simulated from oracle labels with a reviewer-error parameter, not collected from human reviewers. All rules but the judged-grounds rule, a heuristic threshold on a mean score, share one certification engine, which favours the simpler rules by giving them the same statistical guarantee.

The pilot is small. With three classes per configuration, the delegation test is informative only in the few comparisons in which the rules choose different classes: for DSA against track records, 10 of 28 on all data (11 in the common core) under the registered key and 8 of 28 (9) under the reviewed key, and under the registered key most of these comparisons involve a tie at the boundary of the delegated set (A7). Bonferroni correction across classes is conservative. The pilot can show whether the effects exist and how large they are; the confirmatory delegation test needs the second phase.

The environment is synthetic. Documents are short, clean and in English; real files are longer, scanned, multilingual and inconsistent. Control over issuers is signalled only by issuer names. The evidence shift is designed rather than observed, and it is not adversarial: when sources can be corrupted, no procedure that sees the same evidence can avoid deferring some cases to a human (Pandey et al. 2026b). The certification guarantee holds for the calibration generator's distribution; performance under the constructed shift is assessed empirically, and neither says anything about production traffic.

Certification is not authorisation. A certified class tells a human what the agent can be trusted to do; it does not move legal responsibility. Decisions that the law subjects to meaningful human intervention, such as onboarding under AMLR Art. 76(5)(b), and those the policy reserves for humans, stay so however high the measured DSA. We paraphrase public rules but make no claim of regulatory compliance.

9 CONCLUSION

We asked which evidence should decide that an AI agent may take a class of decisions without review, and we tested one answer, the agent's reaction to interventions on the evidence, against a track record and an LLM judge in a prospectively specified pilot with 14 model configurations. The pilot does not show that evidence tests beat a track record. The registered test passed, and the advantage disappeared when one contested intervention arm, the removal of a register excerpt that the policy does not require, was excluded or when the author's review of the oracles' answers was applied; removing the synthetic policy's controller-independence requirement also yielded no statistical support for a DSA advantage across the 30 additional sensitivity variants, which retained the oracles' single-action convention where the exception applies (§8). What held under every key is that an LLM judge reading the agent's explanation passes most of the correct decisions that fail an interventional test, and that a stronger judge does no better; the known-groups check confirmed that the instrument detects a planted memorising agent that a track record would certify.

The lesson we draw concerns method. In an evaluation of this kind the answer key is part of the instrument: it feeds both the test and the criterion against which the test is judged, two separately written oracles can share one contestable reading, and a policy written for a benchmark can be stricter than the law it paraphrases without anyone noticing until the results are in. The key therefore needs the governance a model gets: documented readings, blind review by more than one person, admissible sets where the policy is silent, and a sensitivity analysis over readings next to the primary result (§7.1). We publish the registration, every amendment, the ambiguity log with its legal check, the author's review, the multiverse and the code, so that each step from data to conclusion can be checked.

The second phase follows from this: the policy text aligned with the legal texts and every open term fixed in writing before any agent runs; the facts the oracles use visible to the agents; the key reviewed blind by several raters on a sample drawn where models disagree; the delegation question tested with the sample sizes the pilot's variance implies, on new seeds, under a registration deposited with an external registry before the data; and one model run both bare, through its API, and behind a commercial agent runtime on the same episodes, to test whether the basis measures depend on the wrapping rather than the model. Whether evidence tests earn a place in a delegation rule is, after this pilot, still an open question, and now one that can be asked with an instrument whose answer key has been reviewed.

STATEMENTS

Statement on independence and data: The author is employed at a financial institution. This work was carried out outside that employment, through the author's own practice, Generatywnie; it does not represent the views of the employer and uses no data, processes, systems or other resources of the employer. All environments are synthetic; the counterparty-verification policy paraphrases only public legal texts.

Code and data availability: Generator code, intervention catalogue, baseline implementations, analysis scripts, run logs, the author's review files and the reading notes are released under open licences (Apache-2.0 for code, CC BY 4.0 for texts and data). Every episode, including the held-out and shifted splits, is regenerated deterministically from its seed, and all seeds are published with the code; a private seed range is reserved for the second phase. The repository is https://github.com/speplinski/worktrace (release v1.1; archived at Zenodo under the concept DOI 10.5281/zenodo.23868234, which resolves to the latest version); the results

files the paper cites are in its `paper/pilot/` directory, and every file path in this paper is relative to its root; `results_interim.md` reports all data and `results_core.md` the common core.

Use of AI assistance: This paper was prepared with substantial help from an AI system, Claude (Anthropic), used through the Claude Code agent environment. Its role was as follows. (1) *Literature.* It searched arXiv and the web, read papers and wrote structured reading notes, which are published in the repository; every reference was then checked against arXiv or Crossref metadata where available and otherwise against the publisher's or issuer's page, and the bibliography records how each entry was checked. (2) *Study design.* It proposed and critiqued design options. The author made every design decision and approved the pre-registration and each amendment; the decision log is published. (3) *Code.* The generator, the oracles, the agent harnesses, the LLM judge and the analysis scripts were written by the AI system at the author's direction and reviewed and run by the author. (4) *Writing.* The text was drafted by the AI system from the author's theses, decisions and results; the author revised and approved it and takes responsibility for every claim. The experiments were run by the author on his own computer and API account. *Potential conflict of interest.* The AI system used for this work is developed by Anthropic, whose models are among the configurations evaluated, and an Anthropic model serves as the LLM judge for the two judge-based baselines and for the judge and declared-grounds tests, with a second judge from another developer run as a check (amendment A10). To limit the resulting risk, the analysis plan, its dating and its amendments are described in §5.8; ground truth comes from two deterministic oracles, not from any model; configurations from seven other developers are included; the judge's rubric, its one revision and the trial verdicts are published; and all code, prompts, model outputs and analyses are released. *Funding.* None; no grant was involved. Claude API costs were paid from the monthly API credit included in the author's personal, paid Claude subscription (USD 100 per cycle) and otherwise by the author; OpenAI API costs and local computation were paid by the author. Anthropic had no role in the design, analysis or writing of this paper.

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A MEASURES REGISTERED IN §5.4 THAT §6 DOES NOT REPORT

Calibration differs by class: Basal 2.0-max is overconfident in the discrepancy and freshness classes (ECE 0.26 and 0.27) and calibrated in the onboarding class (0.04), where it always chooses an admissible action (escalate in 46 of 60 episodes, a document request in 14) with a mean confidence of 0.96; Liquid AI d1-3B is worst in the discrepancy class (ECE 0.53: mean confidence 0.84, accuracy 0.35). Liquid AI d1-3B's deferral rule is not monotone: deferring a quarter of its decisions lowers the unjustified rate to 56%, deferring half raises it again to 63%, because its most confident decisions, in the discrepancy class, are mostly wrong. The confidence of the decision models therefore carries usable information for triage, but none of the three is calibrated enough for its stated probability to be read as a risk, which is what a certified threshold rule would need (§6.2).

Table 7 collects these measures; Table 5 reports the calibration of the decision models' confidence and the trade-off of a confidence-based deferral rule. Out-of-permission actions were attempted by no configuration in any class. Protocol incidents (all splits): Gemma 4 26B, 13 reseeded retries after rejected tool calls and one run that hit the turn limit; gpt-oss 120B, one reseeded retry, three decisions recovered from text and one run without a decision; Mistral Small 3.2, six server timeouts and two runs without a decision; none elsewhere. Excluding the three gpt-oss units with text-recovered decisions, the robustness check of §5.7, changes no grant, because the configuration certifies nothing in any class under any rule. Majority voting over the three runs of Claude Haiku 5.5 (all data) certifies the discrepancy class for the DSA rule (upper bound 0.059 against 0.116 with runs as units), leaves the other grants unchanged and leaves the delegation-test selections unchanged (both rules delegate the discrepancy class at $k = 1$ and the discrepancy and onboarding classes at $k = 2$). Escalation calibration varies widely: GPT-6.1 Terra under-escalated 51% of the onboarding-class arms whose only admissible answer was escalation, and gpt-oss 120B over-escalated 37% of the freshness-class arms with an admissible non-escalation answer.

Table 5: Decision models: calibration of the reported confidence against outcome correctness and the deferral trade-off (common core, calibration split, 180 factual decisions each; `paper/pilot/confidence_calibration.md`). ECE is the expected calibration error over ten equal-width bins; Brier is the mean squared error of the confidence as a probability of a correct outcome. "Unjustified" means a wrong outcome or a basis failure under the registered key; the deferral rule sends the given share of decisions with the lowest confidence to a human, the rate is measured among the remaining decisions, and "caught" is the share of all unjustified decisions among the deferred ones. Under the reviewed key the figures change by at most 1.2 percentage points. The 95% bootstrap intervals of the ECE (1,000 resamples) are [0.11, 0.23], [0.17, 0.28] and [0.06, 0.17]; the confidence field is what each model reports, which for Basal equals the probability of the chosen action and for the other two lies below it (ECE on that probability: 0.16, 0.21 and 0.13).

model	mean confidence	accuracy	ECE	Brier	unjustified, no deferral	at 25% deferred	at 50% deferred	caught at 50%
Basal 2.0-max	0.84	0.68	0.158	0.208	61.1%	51.1%	34.4%	71.8%
Liquid AI d1-3B	0.66	0.57	0.203	0.268	66.7%	55.6%	63.3%	52.5%
GPT-6 Luna (Decisions API)	0.72	0.66	0.090	0.184	65.6%	60.0%	52.2%	60.2%

Table 6: The declared-grounds check against the interventions (H2): correct-outcome decisions in the common core, pooled over the judged configurations, under each answer key. The static check is the judge's verdict on whether the documents the agent declared as decisive are those that settle the episode; the interventional result is the reference, so sensitivity and specificity are relative to it (§5.6, §6.2). Source: `paper/pilot/results_core.md`.

key	basis failures flagged	basis failures missed	passes flagged	passes not flagged	sensitivity	specificity
registered	306	168	442	819	64.6%	64.9%
without the contested arms (A1)	160	128	588	859	55.6%	59.4%
reviewed (A13)	198	139	550	848	58.8%	60.7%

Table 7: Calibration and held-out losses, control-arm flips, non-reaction and escalation calibration per configuration and class (common core, registered key; `paper/pilot/appendix_measures.md`, generated from `results_core.md`). Flip rates (null; placebo and sham) are the share of control arms on which the action changed; non-reaction (non-react.) is the share of interventions that made the factual action inadmissible on which the agent kept it; over-escalation (over-esc.) is the share of arms with an admissible non-escalation answer on which the agent escalated, under-escalation (under-esc.) the share of arms whose only admissible answer was escalation on which it did not. "–" marks an arm or measure that does not exist for the configuration or class.

configuration	class	cal. ℓ_out	cal. ℓ_WT	held-out ℓ_out / ℓ_WT	shifted ℓ_out	null flip	placebo, sham flip	non-react.	over-esc.	under-esc.
Basal 2.0-max	discrepancy	40.0%	85.0%	46.7% / 96.7%	26.7%	–	5.0%	81.5%	0.5%	14.3%
Basal 2.0-max	freshness	55.0%	98.3%	43.3% / 96.7%	48.3%	–	13.3%	45.0%	0.0%	–
Basal 2.0-max	onboarding	0.0%	0.0%	0.0% / 0.0%	0.0%	–	0.0%	–	–	0.0%
Claude Haiku 5.5	discrepancy	0.0%	8.3%	0.0% / 13.3%	0.0%	0.0%	0.8%	1.0%	0.0%	0.0%
Claude Haiku 5.5	freshness	10.0%	55.0%	10.0% / 46.7%	25.0%	13.3%	20.0%	1.2%	4.5%	–
Claude Haiku 5.5	onboarding	3.3%	8.3%	6.7% / 10.0%	0.0%	13.3%	10.8%	–	–	4.3%
Claude Opus 5.5	discrepancy	0.0%	10.0%	0.0% / 10.0%	0.0%	0.0%	0.0%	4.1%	0.0%	0.0%
Claude Opus 5.5	freshness	20.0%	41.7%	13.3% / 43.3%	16.7%	3.3%	5.0%	0.0%	0.0%	–
Claude Opus 5.5	onboarding	0.0%	0.0%	0.0% / 0.0%	0.0%	0.0%	0.0%	–	–	0.0%
Claude Sonnet 5.5	discrepancy	0.0%	16.7%	0.0% / 20.0%	1.7%	0.0%	0.8%	4.1%	0.0%	0.0%
Claude Sonnet 5.5	freshness	3.3%	45.0%	6.7% / 30.0%	6.7%	1.7%	3.3%	5.7%	0.0%	–
Claude Sonnet 5.5	onboarding	0.0%	0.0%	0.0% / 0.0%	0.0%	0.0%	0.0%	–	–	0.0%
Bielik 11B v3 (one-pass)	discrepancy	51.7%	95.0%	30.0% / 93.3%	43.3%	–	7.5%	62.0%	5.4%	71.4%
Bielik 11B v3 (one-pass)	freshness	50.0%	98.3%	50.0% / 100.0%	70.0%	–	25.8%	51.0%	0.0%	–
Bielik 11B v3 (one-pass)	onboarding	0.0%	0.0%	0.0% / 0.0%	0.0%	–	0.0%	–	–	0.0%
Gemma 4 26B	discrepancy	5.0%	31.7%	0.0% / 23.3%	1.7%	–	8.3%	8.6%	0.0%	0.0%
Gemma 4 26B	freshness	6.7%	48.3%	0.0% / 33.3%	23.3%	–	7.5%	2.4%	2.9%	–
Gemma 4 26B	onboarding	1.7%	10.0%	3.3% / 6.7%	0.0%	–	5.0%	–	–	4.3%
gpt-oss 120B	discrepancy	20.0%	61.7%	26.7% / 70.0%	23.3%	–	30.8%	10.6%	0.0%	51.0%
gpt-oss 120B	freshness	48.3%	98.3%	36.7% / 100.0%	48.3%	–	38.3%	17.9%	37.1%	–
gpt-oss 120B	onboarding	6.7%	31.7%	13.3% / 33.3%	1.7%	–	29.2%	–	–	15.2%
Mistral Small 3.2	discrepancy	20.0%	75.0%	26.7% / 80.0%	20.0%	–	35.0%	50.7%	1.0%	2.0%
Mistral Small 3.2	freshness	55.0%	95.0%	46.7% / 100.0%	50.0%	–	26.7%	45.2%	0.0%	–
Mistral Small 3.2	onboarding	6.7%	11.7%	0.0% / 10.0%	1.7%	–	18.3%	–	–	5.8%
Qwen 3.8 27B	discrepancy	10.0%	35.0%	13.3% / 43.3%	16.7%	–	9.2%	11.5%	0.0%	0.0%
Qwen 3.8 27B	freshness	0.0%	40.0%	0.0% / 36.7%	26.7%	–	0.0%	3.4%	0.0%	–
Qwen 3.8 27B	onboarding	0.0%	0.0%	0.0% / 0.0%	0.0%	–	0.0%	–	–	0.0%
GPT-6 Luna (Decisions API)	discrepancy	26.7%	70.0%	20.0% / 80.0%	10.0%	0.0%	14.2%	42.6%	0.0%	12.2%
GPT-6 Luna (Decisions API)	freshness	50.0%	91.7%	43.3% / 96.7%	51.7%	0.0%	7.5%	63.3%	0.8%	–
GPT-6 Luna (Decisions API)	onboarding	25.0%	35.0%	23.3% / 26.7%	1.7%	0.0%	28.3%	–	–	19.6%
GPT-6 Luna	discrepancy	8.3%	33.3%	10.0% / 30.0%	10.0%	10.0%	10.0%	6.8%	0.0%	0.0%
GPT-6 Luna	freshness	0.0%	33.3%	0.0% / 26.7%	21.7%	1.7%	0.8%	0.0%	0.0%	–
GPT-6 Luna	onboarding	26.7%	41.7%	13.3% / 33.3%	3.3%	25.0%	27.5%	–	–	24.6%
GPT-6.1 Sol	discrepancy	0.0%	8.3%	0.0% / 10.0%	0.0%	0.0%	0.0%	0.0%	2.5%	0.0%
GPT-6.1 Sol	freshness	0.0%	33.3%	0.0% / 30.0%	25.0%	0.0%	0.0%	0.0%	0.0%	–
GPT-6.1 Sol	onboarding	0.0%	0.0%	0.0% / 0.0%	0.0%	0.0%	0.0%	–	–	0.0%
GPT-6.1 Terra	discrepancy	8.3%	30.0%	13.3% / 30.0%	3.3%	6.7%	15.0%	3.4%	5.0%	0.0%
GPT-6.1 Terra	freshness	0.0%	33.3%	0.0% / 30.0%	16.7%	0.0%	0.0%	0.0%	0.0%	–
GPT-6.1 Terra	onboarding	46.7%	68.3%	60.0% / 66.7%	16.7%	25.0%	28.3%	–	–	51.4%
Liquid AI d1-3B	discrepancy	65.0%	100.0%	53.3% / 100.0%	51.7%	–	0.0%	100.0%	0.0%	100.0%
Liquid AI d1-3B	freshness	63.3%	100.0%	70.0% / 100.0%	73.3%	–	6.7%	87.5%	0.0%	–
Liquid AI d1-3B	onboarding	0.0%	0.0%	0.0% / 0.0%	0.0%	–	0.0%	–	–	0.0%